

Corruption exposure and the stock return: Evidence from U.S. firm operating in Africa

(This draft: August 2020)

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¹ Of the many helpful contributions, the author heartily appreciates the beneficial discussion with Professor K.C. John Wei (Hong Kong Polytechnic University) and the encouragement from Jeffrey Pontiff (Boston College). The author would also like to acknowledge valuable feedback from Kay Giesecke (Stanford University), Han-Hsing Lee (National Chiao Tung University, Taiwan), Chi-Yang Tsou (Hong Kong University of Science and Technology, Hong Kong), Junbiao Yu (Hong Kong Polytechnic University, Hong Kong), Alessandro Zattoni (LUISS University Rome) and seminar participants at the 13th NCTU International Finance Conference, University, Kaohsiung, Taiwan. All errors are the author own's responsibility. Corresponding Fax: (853) 2882-3281.

Abstract

In this paper, we exploit under what circumstances will corruption lead to higher firm values by focusing on the operations of U.S. multinationals in Africa. The result shows that firms with higher corruption exposures are more likely to operate in Africa. Moreover, with higher corruption exposures, U.S. firms operating in Africa experience a higher return related to their peers without African operations. After controlling a range of county- and firm-level variables, we find that high corruption exposure interacting with African operations produces significant positive stock returns. An identification strategy that matches our U.S. sample with Chinese firms with Africa operations confirms our results. Further tests show that our results are robust when we look at different African countries and use alternative corruption measures. Besides, exposure to corrupt countries is not a proxy for state-level corruption within a country. In sum, there is a valuation gain for a multinational if its corruption exposures match the corruption level of operating destinations.

Keywords: Corruption, stock return, international operations, Africa, firm value

JEL: D73, G12, F23, N27, O55, G32

1 Introduction

There has been a large debate on the effect of corruption on firm value. While corruption impedes economic growth of the society,² it may also increase corporate value as a firm pays bribes to increase the probability of winning positive net present value contracts. For instance, Kaikati and Label (1980) and Zeume (2017) show that in the context of multinational operations, paying bribes facilitates doing business in certain countries. In contrast, imposing unilateral anti-corruption regulations that also apply to foreign bribes hurts firm value as it gives a competitive advantage to unregulated foreign peers. Further, Henisz (2000) and Hines (1995) report that foreign bribery payment is necessary in order to protect the position of foreign operations. There are two channels corruption might increase firm value. First, it allows foreign government officials who accept bribes to communicate better and work harder. Second, it enables companies that commit bribery to avoid the administrative delay of foreign bureaucrats (Huntington, 2006; Leff, 2002). This literature suggests that corruption is closely associated with foreign operations.

This paper attempts to shed a light on a hitherto fiercely debated issue in the literature: How does a firm's exposure to corrupt cultures affect its financial returns. On the one hand, corruption reduces firm value because, for instance, bribes reduce cash flow to shareholders.³ On the other hand, however, bribes allow for cutting through the bureaucratic red tape that may bring an even larger countervailing benefit to shareholders in the future. Bad or good, Corruption is inevitable as

² At the society-level, corruption disrupts the allocation of resources and leads to inefficient investments. For instance, Mauro (1995) reports that corruption constitutes a severe obstacle to investment and thus lowers economic growth. Shleifer and Vishny (1993) document that corruption deteriorates social resources allocation by shifting investments from the potentially highest value projects to lowest value projects, and consequently impedes growth.

³ Studies that document the negative effect of corruption include Butler et al. (2009), Dass et al. (2016), DeBacker et al. (2015), Liu (2016), Pantzalis et al. (2008), Smith (2016), Tirole (1996). The discussion of these studies will be in the Literature Review section.

long as the government intervenes in the market and plays a part in the economic world (Acemoglu & Verdier, 2000; Ehrlich & Lui, 1999; Rose-Ackerman, 2013).

Built upon the existing studies, this paper asks a specific question: Under what circumstances will a firm's exposure to corrupt countries lead to higher financial returns? Firms operating internationally are prone to the institutional environment of foreign countries. As they need to deal with different foreign governments, this increases the sources of corruption.⁴ As empirical evidence, O'Donovan et al. (2019) show it is highly pervasive that firms use offshore financial activities to pay bribes and win business contracts from government agents of corrupt countries. Using data from the Panama Papers, they disclose that 214,000 shell companies have been established in tax havens in the past half-century and many of these companies are fully connected with subsidiaries of publicly traded firms to finance bribe payments.

To answer the proposed research question, we conjecture that a firm with a higher level of corruption exposure would gain benefits if the firm operates in a highly corrupt region. The difficulty is, nonetheless, to capture the degree of corruption exposure of a firm and to identify the level of corruption of a region are both challenging. Because corruption is largely unobservable. For instance, although the Department of Justice (DOJ) has imposed U.S. Foreign Corrupt Practices Act (FCPA) since 1977, only 143 corruption actions were enforced against publicly listed

⁴ Any intervention may affect the optimal allocation of resources by governments. Given the fact that governments play an important part in the economic world, the efficiency of their resource allocation depends on the difficulty of social monitoring. In the case that the rewards (wages) for government intervention are lower relative to the increased difficulty of social monitoring, they would have incentives to seek additional rents (Acemoglu & Verdier, 2000). While might at a different degree, corruption exists in both developed and emerging markets. In the developed market, for example, Amore and Bennedsen (2013) find corporate rent-seeking activities in Denmark, Duggan and Levitt (2002) find evidence of corruption in Japan, and Bobonis et al. (2016) discuss monitoring corrupt activities in Puerto Rico. In the emerging market, see corrupt evidence of Callen and Long (2015) for Afghanistan, Ferraz and Finan (2011) for Brazil, Olken (2007) for Indonesia, Oliva (2015) for Mexico, Sequeira (2016) for Mozambique and South Africa, and Svensson (2003) for Uganda.

U.S. firms according to FCPA between 1978 to 2013 by DOJ and the Securities and Exchange Commission (SEC) (Karpoff, Lee, & Martin, 2017; Zeume, 2017). One way to address this issue, as suggested by Zeume (2017) is to examine exogenous benefits and costs of doing business in perceptively corrupt countries. Following this line of thoughts, we employ a quasi-experimental design that allows us to study market returns of firms that are subject to a possible exogenous increase in doing business in perceptively corrupt countries and relate to their corruption exposure to their operations in African countries.

Exploiting the operations in Africa is appropriate only if corruption exposure had a substantial effect on firms. We argue that this is the case for at least three reasons. In the first place, corruption is so pervasive in Africa to which no other continent would compare. According to Transparency International's 2018 Corruption Perceptions Index, Africa remains the most corrupt continent in the world.⁵ It reveals that regional Sub-Saharan Africa is the lowest (32 out of 100). While five countries in Northern Africa performed relatively better than the Sub-Saharan African countries, their average score of 34.6 is still lower than any other continents in the world.⁶ Further, a recently published report by Transparency International discloses that one in four people in Africa pay bribes,⁷ meaning that approximately 130 million people pay bribes. The number also suggests a 73% increase in Africans who pay bribes compared to the figure in the earlier version's report.⁸

⁵ Transparency International ranks 180 countries and territories by their perceived levels of public sector corruption, which is evaluated using scores from 0 (the highest corrupt) to 100 (the lowest corruption). See https://www.transparency.org/files/content/pages/2018_CPI_Executive_Summary.pdf.

⁶ The five Northern African countries (the Corruption Perceptions Index) are Morocco (43), Tunisia (43), Algeria (35), Egypt (35), Libya (17). Transparency International reports the combined score for five Northern African countries with Middle East countries, the combined score is 39. For other regions, the average score for Eastern Europe & Central Asia is 35, for Americas is 44, for Asia Pacific is 44, and for Western Europe & European Union is 66. The most corrupt country is Somalia that is ranked 180 with a score of 10.

⁷ https://www.transparency.org/news/pressrelease/one_in_four_people_in_africa_pay_bribes_survey_says

⁸ https://www.transparency.org/news/feature/corruption_in_africa_75_million_people_pay_bribes. The title of the report is "People and Corruption: Africa Survey 2015"

Second, the United States has increased its operations in Africa intensively and is foreseeable to continue doing so in the future. This century has witnessed that the United States has increased its participation in the African market under the African Growth and Opportunity Act (AGOA). In 2017, the United States has a \$39 billion goods trade with Sub-Saharan African countries and a \$10.8 billion goods trade deficit, increased from \$4.1 billion deficit in 2016 due to increase in oil imports. According to the Office of the United States Trade Representative, the U.S. Congress extended AGOA to 2025,⁹ indicating an anticipated increase in trade growth between the United States and Africa.

Third, the U.S. firms have meanwhile extensively increased global operations, which involve countries with varying degrees of corruption. The recent dispute about the globalization strategy of U.S. multinationals makes it critical to think about how globalization benefits their investors. While American investors are averse to corruption in foreign countries (Wei, 2000), there is ample empirical evidence that corruption is substantial in the United States.¹⁰ To a certain extent, the competition depends on who is more familiar with the corrupt culture. Given that all of the African countries are perceived more corrupt than the United States, operating in Africa gives those U.S. firms with a higher degree of corruption exposure an advantage over their U.S. peers with a lower degree of corruption exposure.¹¹

⁹ See <http://ustr.gov/countries-regions/africa>

¹⁰ See, for example, Borisov et al. (2015), Butler et al. (2009), Campante and Do (2014), Dass et al. (2016), Liu (2016), Pantzalis et al. (2008), and Smith (2016). According to the 2018 Corruption Perceptions Index (*CPI*) of Transparency International and the 2017 Control of Corruption Index (*CCI*) of the World Bank, the United States is ranked at the 22nd and 24th places in the world, respectively.

¹¹ The African country with the highest ranks in both *CPI* and *CCI* are below the ranking of the United States. With *CPI*, Seychelles has the highest score among African countries and is ranked at the 28th place in the world. With *CCI*, Botswana has the highest score among African countries and is ranked the 45th place in the world.

To examine the effect of corruption exposure on firm value, we focus on publicly listed U.S. firms. We measure a firm's corruption exposure by using a similar approach to Zeume (2017) and constructing the variable that captures firms' exposure to high-corruption countries using a combination of firm-level subsidiary locations, which in our main analyses is based on Transparency International's *CPI* and the World Bank's *CCI*. With this measure, we first investigate firms' propensity to operate in Africa. Both subsidiary- and firm-level analyses show that corruption exposure is positively associated with a firm's propensity to operate in Africa. At the subsidiary-level, after controlling a range of country-level features, the result shows that the higher the corruption exposure, the greater the likelihood to have subsidiaries in African countries. At the firm-level, we further control a set of firm-level factors and find that corruption exposure is significantly and positively related to the proportion of a firm's African subsidiaries to the total number of the firm's subsidiaries.

In our main analyses, we explore how corruption exposure and African operations affect firm value. We document that, on average, firms with high corruption exposure and operating in Africa experience a positive stock return. This return is significantly higher than the average return of firms without African operations, which is shown to be negative. Moreover, we create annually rebalanced corruption exposure weighted portfolios. We find that the portfolios without African operations have negative abnormal returns, whereas the portfolios with African operations offset the negative effect and earn positive abnormal returns. For example, a portfolio of firms without African operations weighted by either *CPI* or *CCI* has a statistically significant Fama and French's (2015)'s five-factor-plus-momentum-model abnormal daily return of -0.3 basis points. Comparatively, a portfolio of firms with African operations earns an abnormal daily return of 0.5 basis points. Further, multivariate analysis reveals that, after controlling for other market-, firm-,

country-level variables and institutional factors that potentially affect firm value, the interaction of high corruption exposure with African operations is associated with a significant positive abnormal return. We thus conclude that corruption facilitates doing business in Africa, which is in turn associated with an increase in value for the average firm.

Both stock returns and decisions of operating in Africa may be driven by conditions and factors specific to the United States that are not directly observable. This is also related to the challenge that we mentioned earlier that corruption itself is usually unobservable. If this is the case, then endogeneity may arise from omitted variables and could be an important issue to affect our results. To address this concern, we adopt an identification strategy by using propensity score matching approaches, where the control group is a set of Chinese firms that are also significantly involved in African operations. The factors that drive U.S. and Chinese firms to operate in Africa are exogenous to each other. Further, the stock returns of Chinese firms that operate in Africa are less likely to be abnormal due to the proximity of the degrees of corruption between China and African countries. Further, as the reference is outside of the United States, the result after matching identifies and reflects the effect of U.S. firms' African operations. If endogeneity drives our results, then there would be a change in our results and firms with high corruption may show a lower abnormal return than firms with low corruption. Using propensity score matching techniques, we document that firms with high corruption have a consistently stronger effect than firms with low corruption in terms of both economic and statistical significance. In essence, this finding suggests that the African operations associate with corruption exposures to affect stock returns are not driven by endogeneity.

We also examine whether the results would change if we look at different African countries. While Africa is the most corrupt continent as a whole, the degree of corruption varies across African

countries. If foreign bribery payments facilitate U.S. firms to do business in African countries with a relatively higher degree of corruption, then firms with low corruption exposure would not so valuable would still contribute to firm value and thus be positively related to stock returns. To test whether or not this is the case, we carry out the subsidiary-level analysis by focusing on those firms with subsidiaries in Africa. We control for corruption distance between the United States and each of the African countries that host our sample subsidiaries along with other differences in country-level factors. The results show that although corruption distance is negatively related to stock returns, there is a consistently significant and positive effect of corruption exposure on firm value. Therefore, our firm-level results are not driven by differences across African countries.

The empirical tests on corruption exposure might subject to the measure of corruption. We therefore consider alternative measures of corruption that are validated by Wilhelm (2002) — the index of the overabundance of regulation or unnecessary restriction of business (*Excess Regulation*) activity and the index of black market activity (*Blank Market*). Notably, the results with these two alternative corruption measures are highly consistent with the findings that we have documented. Further, one might suspect that our measure of corruption exposure is just a proxy for state-level corruption, which is shown to have a significant effect on financial outcomes and activities in the literature (Butler, Fauver, & Mortal, 2009; Dass, Nanda, & Xiao, 2016; Smith, 2016). Nevertheless, we find that state-level corruption does not proxy for foreign corruption, and in fact, the co-effect of state-level corruption and African operations negatively affect firm value.

We conduct a range of robustness checks. First, after controlling different and additional variables for size, cultural distance, momentum factor, and firm age, our results do not materially change. Further, we exclude smaller and larger sample firms, the results still hold. Another potential influence effect is the global financial crisis that significantly disrupts the stock market. Indeed,

our results show that the co-effect of corruption exposure and African operations on stock returns is significant in the subsample period before and after the global financial crisis but is insignificant during the subsample period of the global financial crisis, suggesting that the effect of corruption exposure takes place during the “normal time”. Last but not least, using the change in return on capital (*ROC*) as the dependent variable, we find that the co-effect of corruption exposure and African operations is positively related to changes in *ROC*. This finding suggests that the effect of corruption exposure and African operations on firm value is associated with fundamental changes.

A key contribution of this paper is to provide firm-level evidence of corruption exposure impact on firm operations and value. Very few studies have provided similar evidence, and most relevant study to ours is the work by Zeume (2017). We extend Zeume’s (2017) work in several ways. First, Zeume (2017) adopt an event study method, whereas we employ several different techniques, including univariate test, portfolio analysis, multivariate regression, and propensity score matching. With these different methods from Zeume (2017), we achieve a similar result to Zeume (2017), showing that corruption exposure can positively affect firm value. Second, Zeume (2017) focuses on the United Kingdom, whereas we provide supporting evidence from the United States. Third, more importantly, we examine the effect of firm operations in Africa. In recent years, Africa has become the focus of investments from countries outside of the continent.¹² As a result, although the worldwide foreign direct investments are continuously down at the same time, the popularity of investments in Africa does not diminish.¹³ In the consideration of both high levels of corruption

¹² According to figures from the United Nations, Africa received \$46 billion in foreign direct investments (FDI) in 2018, up by 11% from 2017. This increase is, to a large extent, attributed to the competition between the United States under Donald Trump’s and other major global players such as China’s (Forde, 2019).

¹³ See United Nations Conference on Trade and Development UNCTAD’s World Investment Report 2019: https://unctad.org/en/PublicationsLibrary/wir2019_en.pdf

and growth potential of Africa, our important contribution is to establish a basic stylized fact regarding the financial consequence of operating in one of the most corruption areas.

In this study, we provide some evidence on multinational operation studies by examining corruption, one of the most important country-level exposures. Especially, we provide evidence concerning the effect of country-level exposure on corporate value. The existing literature investigating the effects of multinational operations focuses mainly on intangible assets, taxation and cash holding (T. Gu, 2017; Hanlon, Lester, & Verdi, 2015; Harford, Wang, & Zhang, 2017; Huizinga, Laeven, & Nicodeme, 2008; Morck & Yeung, 1991; Pinkowitz, Stulz, & Williamson, 2012; Piotroski, De Simone, & Tomy, 2018). While there have been several studies on the effect of country-level factors (Denis, Denis, & Yost, 2002; Desai, Fritz Foley, & Hines Jr, 2008; Henisz, 2000; La Porta, Lopez-de-Silanes, Shleifer, & Vishny, 1999; Slangen & Beugelsdijk, 2010; Xu & Shenkar, 2002). Only a handful of studies noticed the effect of corruption (O'Donovan et al., 2019; Pantzalis, Park, & Sutton, 2008; Zeume, 2017). Therefore, by which way informal institutional factors, such as corruption, harm or benefit multinational companies remained an open question in the finance and economic research. The findings in this paper provide a novel angle to look at the effect of multinational operations, and especially meaningful for those in Africa.

The remainder of the paper is structured as follows. Section 2 reviews the literature. Section 3 describes the data. Section 4 presents the empirical results, and Section 5 concludes the paper.

2 Related literature

In this section, we review the literature on the effect of corruption. We first define what corruption is, then we look at why corruption exists extensively in Africa, and finally we discuss how corruption affects corporate financial practices and values.

2.1 The definition of corruption

Shleifer and Vishny (1993) define corruption as “the sale by government officials property for personal gain”. According to this definition, government officials have discretion over the provision of properties owned by the state, such as licenses and permits, and they can collect bribes through this discretion by charging private agents.¹⁴ Ehrlich and Lui (1999) claim that corrupt behavior is to exercise the chance to obtain bribery, which is the side payment to close the gap between free-market prices and shadow prices generated due to government intervention in the economy by taking responsibility of resource allocation in the economy. Acemoglu and Verdier (2000) argue that corruption is “an unpleasant side effect” as a result of government intervention designed to direct financial activities and correct market failures. These definitions are highly consistent with Rose-Ackerman (2013) who suggest that corruption is inevitable when the government acts as a market player.

The international operation increases information asymmetries between a company and foreign governments because information concerning counterparts is incomplete. In the world with incomplete information, there are always rent-seeking opportunities.¹⁵ The company may need to pay a premium to access a new institutional environment. These extra costs are also known as the liability of foreignness (Calhoun, 2002; Denk, Kaufmann, & Roesch, 2012; Zaheer, 1995). Meanwhile, the arrival of foreign companies increases the difficulty of monitoring, which may

¹⁴ The properties that create opportunities for collecting bribes do not necessarily to be physical goods. For example, they can be the rights to get access to certain areas or buy certain land with a fixed price, or the actions that prohibit the entry of competitors.

¹⁵ In an ideal economic world, the market is informational complete, and the government is independent of the market and is rational. Under these conditions, governments do not have to intervene the market. However, the literature has documented that the market is incomplete, and the government is not independent of the market, this leads to the necessity of government intervention. Meanwhile, studies have also suggested that the government is not rational. The evidence of the collapse of former communist countries demonstrates that the central resource allocation by the government could be problematic.

generate extra costs for the governments as well. Governments need to hire bureaucrats to perform duties on their behalf, which may add another layer of incomplete information because of information asymmetry between the government and bureaucrats. These bureaucrats may also be subject to “bounded rationality” when collecting information, making decisions, and implementing policies (Acemoglu & Verdier, 2000). Moreover, these bureaucrats as self-interest agents have superior information, which leaves room for them to seek-rent. This is likely because bribes are largely unobserved. Using evidence from the Panama Papers, O’Donovan et al. (2019) report it is widespread for companies around the world to use foreign subsidiaries to finance corruption activities. Karpoff et al. (2017) suggest that foreign bribery is not easy to be caught even in the countries where regulations are strictly enforced such as the United States. Because both companies and bureaucrats tend to hide bribery activities, two layers of information asymmetries would leave big rooms for the bureaucrats to have a strong incentive to use their informational advantage associated with their positions to seek additional rents.

2.2 Corruption in Africa

Corruption is ubiquitous in Africa. Svensson (2003) documents that prices of public services in African countries are to a certain extent determined in order to extract bribes. In a study on health care industry, N’Guessan (2004) reports that because information asymmetry between foreign companies and African governments and agency problems between the government and bureaucrats, foreign pharmaceutical laboratories and health care providers can bribe bureaucrats to hide their true costs of health cares. Due to the lack of information about the true values of international pharmaceutical services, the corrupt behavior of bureaucrats would cause sick people to be unable to bear the costs of receiving adequate treatments.

Why corruption is so pervasive in African? D'Aoust and Sterck (2016) attribute corruption in Africa to the judge's judicial discretionary power. Due to the multiplicity and overlap of the judicial system in African countries, the system used to make judgment can be chosen. This encourages corruption as the judge may choose the system for their private benefits instead of retributive principles. The related literature shows that in Africa, even if small bribes would take effect on administrative actions in tax evasion and tariff reduction, which are two major driving factors of the engagement of firms in corruption (Delavallade, 2012; Sequeira, 2016). Contrary to the literature that corruption negatively affects economic growth, Voors et al. (2011) find that in African countries, a higher income stimulates corruption. Arezki and Gylfason (2013) provide supporting evidence showing that for Sub-Saharan African countries, resource rents constitute a significant source of income. However, higher resource rents are also associated with more corruption.

2.3 The financial consequence of corruption

While literature documents a negative effect of corruption on the economic growth at the macroeconomic level,¹⁶ the effect of corruption at the firm-level is still under fervent debate. This section reviews empirical findings on how corruption affects corporate finance.

2.3.1 *The effect of corruption on corporate practices*

The aim of companies to deal with governments is to obtain licenses, permits, rights, actions, or other properties that controlled by the government, it is therefore financially indifferent to them to pay the state or pay government officials personally. According to Ehrlich and Lui (1999), paying

¹⁶ The studies argue that corruption is associated with malfunctioning government institutions, which lead to adverse financial and economic consequences (Mauro, 1995; Shleifer & Vishny, 1993). Even so, scholars also suggest that corruption is inevitable as long as governments intervene in the economy (Acemoglu & Verdier, 2000; Rose-Ackerman, 2013).

bribes for a firm serves a similar function to other investment activities. The only difference is that it is the investment in political capital instead of physical capital or human capital. They argue that corruption does not have net costs as it only involves transfer monetary items from bribe payer to bribe receiver. To a certain extent, bribery can ameliorate information and transaction costs associated with incomplete markets by directing scarce resources to entrepreneurs with the gumption who usually are higher bidders. In addition, Huntington (2006) and Leff (2002) argue that it enables entrepreneurs to overcome bureaucratic delay and cumbersome regulations, empirical evidence stresses the costs of corruption.

The empirical findings concerning the effect of corruption on corporate practices, however, is not favorable. Liu (2016) reports that corrupt firms tend to engage in earnings management, accounting fraud, option backdating, and opportunistic insider trading. DeBacker et al. (2015) find that firms with great exposure to corruption culture evade more tax. Further, Tirole (1996) documents that disclosed corruption affects corporate reputation, causing a firm to suffer from it long even after it is gone. Using state-level data, Butler et al. (2009) show that corruption is positively associated with credit risks and bond yields. Smith (2016) finds that firms located in corrupt areas tend to hold less cash and use higher leverage.

2.3.2 The effect of corruption on firm value

How exposure to corrupt culture does affects the financial outcome? The literature has not yet reached a unanimous agreement too. There have been a handful of study documents a negative effect of corruption. For instance, Dass et al. (2016) report that firms located in more corrupt areas have a lower value. They argue that corrupt environment induces information asymmetry, in that firms in more corrupt areas are more likely to disclose less information by bribing to government officials. Or, they may bribe in order to hide certain information. It is also likely that corrupt

officials threaten firms to exact bribery, firms therefore tend to shield themselves by becoming opaquer, which would eventually reduce firm value. In foreign operations, especially when firms are active in different markets, the effect of corruption can be even more prominent. Due to its illegal nature, contracts obtained through bribery are associated with higher political risk as well. Therefore, in some countries, the costs of bribery can be substantial (Pantzalis et al., 2008).

Several studies, in contrast, find a positive valuation effect of corruption. For instance, Borisov et al. (2015) report that corruption entangles with lobby, which has shown to contribute to firm value. They suggest that there are two channels through which lobbying adds value to the firm — it influences the actions of legislators to make to encourage favorable policies and it reinforces communication with policymakers. The motivation behind the lobbying and bribery is to establish a political connection and to accumulate political capital. Amore and Bennedsen (2013) find that political connection increases firm value, even in a perceived low corruption country such as Denmark. Cooper et al. (2010) also show that corporate political contribution is positively related to future stock returns. In the international setting, Zeume (2017) argues that bribes increase the firm competitive advantage over foreign peers and facilitate doing business in foreign countries. The author reports that imposing unilateral anti-bribery regulations domestically reduces the firm value. Using the Panama Papers, O'Donovan et al (2019) disclose that it is fairly common for public listed companies using offshore vehicles to finance bribe payments and corruption activities. In addition, Desai et al., (2007), Hines (1995) and Kaikati and Label (1980) indicate foreign bribery may benefit outside shareholders as it helps companies to avoid governmental claims on corporate cash flows such as tax and tariff payments.

Taken together, corruption may affect firm value in either a positive or a negative way. While the results in the literature are mixed, zero corruption does not exist.¹⁷ Therefore, corruption is an important issue that is worthwhile to deeply investigate. In this paper, we revisit this issue by focusing our attention on Africa, one of the most corrupt environments.

3 Data description

3.1 Corruption exposure index

The corruption exposure index that we construct is based on two data sources. The first data source is Transparency International's Corruption Perceptions Index (*CPI*), which defines corruption as "the misuse of public power for private benefit". This definition is in line with the definitions of corruption that we discussed in the Literature Review section. Transparency International has published *CPI* since 1995 and ranked 176 countries by "their perceived levels of public sector corruption on a scale from 100 (very clean) to 0 (highly corrupt), as determined by expert assessments and opinion surveys."¹⁸ The use of *CPI* as the measure of corruption is well-accepted in the finance literature (DeBacker et al., 2015; Liu, 2016; O'Donovan et al., 2019; Pantzalis et al., 2008; Zeume, 2017).

The second data source to construct the corruption exposure index is the World Bank's the Control of Corruption Index (*CCI*) that captures "perceptions of the extent to which public power is

¹⁷ Zero corruption suggests an absolute rational or zero-intervention of a government. The only difference in corruption across countries is that less developed countries are hard to rationalize their government intervention (Acemoglu & Verdier, 2000; Rose-Ackerman, 2013).

¹⁸ See <https://www.transparency.org/research/cpi/overview>. After 2012, Transparency International ranges *CPI* between 100 (very clean) and 0 (highly corrupt). We adjust this range to between 10 and 0 with divided the post-2012 *CPI* by 10. According to Transparency International, the least corrupt African country is Botswana with a *CPI* score of 5.95 and the most corrupt African country is Somalia with a *CPI* score of 1.12. The *CPI* of the United States is 7.33. The detailed *CPI* scores are reported in [Appendix I.A](#).

exercised for private gain, including both petty and grand forms of corruption, as well as capture of the state by elites and private interests.”¹⁹ *CCI* is one of the six components of Worldwide Governance Indicators (*WGI*), which is produced by Kaufmann et al. (2011). *CCI* estimate gives “the country’s score on the aggregate indicator, in units of standard normal distribution, i.e. ranging from approximately -2.5 (highly corrupt) to 2.5 (very clean)”.

To construct the measures of the overall exposure of a firm to corruption, we use a similar approach to Zeume (2017) who estimate corruption exposure by the sum of the weighted country-level *CPI* as follows,

$$CEI_CPI_{i,t} = \sum_{c \in C} [(10 - CPI_c) \times \frac{NS_{i,c,t}}{TNS_{i,t}}] \quad (1)$$

where $CEI_CPI_{i,t}$ is corruption exposure index constructed according to Transparency International’ *CPI* of firm i in year t , CPI_c is the average Transparency International’s Corruption Perceptions Index of country c over the sample period, $NS_{i,c,t}$ is the number of subsidiaries in country c owned by firm i in year t , $TNS_{i,t}$ is the total number of subsidiaries of firm i in year t .

Using the same approach, we construct an alternative measure to estimate corruption exposure by the sum of the weighted country-level *CCI* as follows,

$$CEI_CCI_{i,t} = \sum_{c \in C} [(2.5 - CCI_c) \times \frac{NS_{i,c,t}}{TNS_{i,t}}] \quad (2)$$

¹⁹ See <https://datacatalog.worldbank.org/control-corruption-estimate-0>. According to the World Bank, Somalia is the most corrupt country in the world (therefore is also the most corrupt country in Africa) with *CCI* of -1.71. The least corrupt African country is Botswana with *CCI* of 0.98. The *CCI* of United States is 1.41. The detailed *CPI* scores are reported in [Appendix I.B](#).

where $CEI_CCI_{i,t}$ is corruption exposure index computed according to the World Banks's CCI of firm i in year t , CCI_c is the average the World Bank's Control of Corruption Index of country c over the sample period, $NS_{i,c,t}$ is the number of subsidiaries in country c owned by firm i in year t , $TNS_{i,t}$ is the total number of subsidiaries of firm i in year t .

By construction, both measures are increasing in firms' exposure to corruption. In terms of CPI , country corruption values are bounded between 0.65 and 9.15 because $10 - CPI$ is $10 - 0.35 = 0.65$ for the least corrupt country (New Zealand) and is $10 - 0.85 = 9.15$ for the most corrupt country (North Korea). In terms of CCI , country corruption values are bounded between 0.11 and 4.21 because $10 - CCI$ is $10 - 2.39 = 0.11$ for the least corrupt country (Denmark) and is $10 - (-1.71) = 4.21$ for the most corrupt country (Somalia). In both corruption exposure measures, we assume that countries with a greater number of subsidiaries are more important in affecting firm value.

3.2 Corruption and stock data

To numerically capture the degree of corruption exposure using the above measures, we collect a sample of U.S. listed firms from the Orbis database for the period 2004 to 2014. An important merit of the Orbis database is that it shows all foreign subsidiaries of listed firms and contains the information regarding the country of incorporation for each subsidiary. This allows us to identify the corresponding degree of corruption associated with each subsidiary. We merge Orbis dataset with the CEPII (Centre d'Etudes Prospectives et d'Informations Internationales) dataset and build a comprehensive database relating to country-level information, including continent to which a country belongs, geographical distance between host and home countries, and official language of the country where the foreign subsidiaries are located. In line with the literature, we exclude financial and utility firms. At the subsidiary level, we first drop observations without country

information and ensure that each subsidiary is controlled by the firm.²⁰ After this sorting, we have 727,028 observations that come from 253,479 subsidiaries of 3,865 firms.

[\[Table 1\]](#)

We collect stock returns data from Datastream. Similar to Chui et al. (2010), we include all common stocks that are listed on the major exchanges in the United States (NYSE, NASDAQ, and AMEX) and exclude foreign country sample for cross-listed stocks. Further, if a stock has multiple share classes, we only include its primary class in our sample. To alleviate the survival bias in our analyses, we include stocks from the "Dead" stock list as provided by Datastream. One issue with Datastream stock data, as pointed out by Chui et al. (2010), is that a stock return of zero may be a result of no trading. To remedy this issue, we follow Chui et al. (2010) and calculate a stock's return only if the trading volume of this stock is positive in the current as well as previous months. The trading volume data is also collected from Datastream.

[\[Table 2\]](#)

[Table 2](#) reports the summary statistics for the variables included in our study. *Mean return* is computed as the geometric mean of the daily return after subtracting risk-free rate. *Abnormal return* is calculated as the geometric compound daily abnormal returns after matching a firm's stock return with Fama and French's (2015) five factors as well as the momentum factor (the average return on the two high prior return portfolios minus the average return on the two low prior return portfolios). The mean (median) values for *Mean return* and *Abnormal return* are 0.77 (0.66) and 0.21 (0.02), respectively. The mean (median) values for *CEI_CPI* and *CEI_CCI* are

²⁰ To ensure that a firm assumes control over its subsidiaries, we require that it has at least 50.01% of ownership.

2.72 (2.65) and 1.10 (1.07), respectively. In addition to the corruption exposure index, we explore a large number of firm-level and country-level variables that may explain the variation in stock returns. These variables will be discussed in later relevant sections. Their summary statistics are also shown in [Table 2](#) and more detailed description is in [Appendix II](#).

4 Results

In this section, we first report the results on whether corruption exposure relates to U.S. firms' African operations, and then investigate how it affects firm value through stock returns.

4.1 Corruption exposure and operations in Africa

To check the validity of corruption exposure indices that we constructed using the Corruption Perception Index (*CPI*) of Transparent International and the Control of Corruption Index (*CCI*) of the World Bank, we examine the extent to which corruption exposure is related to the possibility of its operations in Africa. Our motivation for these validity checks is based on both the practical observation and theoretical literature. As discussed earlier, Africa to date is the most corrupt continent. The causes of corruption are complicated and usually rooted in the tradition and culture in the specific region (Treisman, 2000; You & Khagram, 2005).²¹ Therefore, Africa provides an excellent platform for us to observe the association of a firm's corruption exposure with the local corrupt environment. To test whether this relation holds, we conduct both subsidiary-level and firm-level analyses.

4.1.1 *Subsidiary-level analysis*

²¹ For example, both New Zealand and African countries have a higher degree of ethnic diversity. The former is the least corrupt country in the world but the latter is the most corrupt region (Easterly & Levine, 1997). The purpose of our study is not to investigate the causes of corruption in Africa.

At the subsidiary-level, to investigate the relation between corruption and the operation in Africa, we follow Siegel et al. (2011) and estimate the following regression:²²

$$\begin{aligned}
AfricaSub_{j,i,t} = & \beta_0 + \beta_1 CEI_{i,t} + \beta_1 EgaDis_{j,i} + \beta_2 DifLaw_{j,i} + \beta_3 ComLan_{j,i} \\
& + \beta_4 ComRel_{j,i} + \beta_5 GeoDis_{j,i} + \beta_6 DifTax_{j,i} + \beta_7 MinPro_{j,i} \\
& + \beta_8 UneBen_{j,i} + \varepsilon_{j,i,t}
\end{aligned} \tag{3}$$

where the subscripts j, i, t represent country j where a subsidiary of U.S. firm i is located in year t . The dependent variable (*AfricaSub*) is a dummy variable that equals one if the subsidiary is located in Africa and zero otherwise. The key interest of the independent variable is the corruption exposure index (*CEI*), as measured by *CEI_CPI* and *CEI_CCI* as defined in Section 3.1. We estimate a probit regression of whether a subsidiary is located in Africa. Since this is essentially the country-level analysis, we cluster the residuals on the country to compute the t -statistics for the estimated coefficients.

In addition to corruption exposure index, we control a range of country-level factors that affect cross-border activities as defined in Siegel et al. (2011). For example, according to Siegel et al. (2011), the distance in the degree of egalitarianism between two countries significantly and negatively affects cross-border activities between the two countries, we follow their study and include the egalitarianism distance (*EgaDis*) from Schwartz (2014) to measure a country's degree

²² Siegel et al. (2011) document that the distance in corruption between the two countries is negatively related to their bilateral activities. We have three main differences from Siegel et al. (2011). First, we investigate the operations of U.S. firms in Africa. Our study therefore focuses on one-way instead of two-way cross-border financial activities. Second, we look at the operations of foreign subsidiaries of U.S. firms, rather than cross-border activities between two independent identities. Third, we focus on Africa where all countries are ranked below the United States in terms of corruption, whereas Siegel et al. (2011) include countries either more corrupt or less corrupt than the United States. Therefore, U.S. firms with a higher degree of corruption exposure are likely more familiar with corruption culture; in other words, more corrupt firms have a shorter corruption distance from Africa, and therefore they are expected to be more likely to operate in Africa.

of egalitarianism. Further, law, language, and religion are widely cited country-level factors affecting finance activities (Demirgüç-Kunt & Maksimovic, 1998; Grinblatt & Keloharju, 2001; Hilary & Hui, 2009; La Porta, Lopez-de-Silanes, Shleifer, & Vishny, 1998; Spolaore & Wacziarg, 2016; Stulz & Williamson, 2003). For instance, law, language, and religion differences, as well as geographical distance, could be barriers impeding cross-border transactions. We therefore follow Siegel et al. (2011) and control these effects by constructing the following variables: Different law family (*DifLaw*), common language (*ComLan*), and common religion (*ComRel*). We obtain language data from CEPII, and law and religion data from Djankov et al. (2007). Geographic distance (*GeoDis*) is captured by calculating a thousand kilometers between the biggest cities in two countries. The data for *GeoDis* are obtained from CEPII. While it is intuitive that geographical distance obstructs cross-border activities, other institutional factors can matter even more for financial development and activities (Rodrik, Subramanian, & Trebbi, 2004). Therefore, we follow Siegel et al. (2011) and include corporate tax rate difference (*DifTax*), minority investor protection (*MinPro*), and unemployment benefits (*UneBen*) to control institutional effects. Tax data are obtained from the World Bank, data on minority investor protection are from The Doing Business project developed by the World Bank, and unemployment benefits are from Botero et al. (2004)

[\[Table 3\]](#)

Panel A in Table 3 reports the regression results. The estimated coefficients of *CEI_CPI* and *CEI_CCI* are positive and significant throughout all regressions at the 5% level or better, which is consistent with the idea that firms with a higher degree of corruption exposures tend to have operations in African countries. This result holds when we control only for GDP of the local country (columns 1 and 5), when we further control for egalitarianism distance (columns 2 and 6), when we further control for law, language, religion differences and geographical distance

(columns 3 and 7), and when we further control for differences in tax rates, minority shareholder protections and unemployment benefits between the United States and host countries (columns 4 and 8), in addition to corruption exposure. The results also show that egalitarianism distance is negatively related to the possibility of the operations in Africa, this is consistent with the findings in Siegel et al. (2011). Moreover, common language and religion are positively related to the possibility of the operations in Africa, whereas the differences in tax rate and minority investor protections are negatively related to the possibility of the operations in Africa. It is worth noting that, the adjusted R^2 in columns 4 and 8 is as high as 70%, suggesting these variables reported in Siegel et al. (2011) has a relatively high power to explain the possibility of cross-border operations of U.S. firms in African countries.

4.1.2 Firm-level analysis

At the firm-level, we examine how corruption exposure affects the degree of operations in Africa as measured by the percentage of a firm's subsidiaries in Africa to the firm's total number of subsidiaries. To ensure the firm-level test is unbiased, we first need to determine whether the firm has operations in Africa; and then if it has, the degree of the operations. We therefore run a tobit regression with controlling firm-level variables as those in Zeume (2017), in addition to the country-level variables as the above but aggregated to the firm-level.

These results are reported in Panel B of [Table 3](#), where *CEI_CPI* is used as the measure of corruption exposure from columns (1) to (4) and *CEI_CCI* is used in columns (5) to (8). In columns (1) and (5), we control only for firm size; in columns (2) and (6), we further control for other firm characteristics; in columns (3) and (7), we further control for egalitarianism distance, law, language, religion and geographical distance; and in columns (4) and (8), we further control for differences in tax rates, minority shareholder protections and unemployment benefits, in addition

to corruption exposure. Throughout all regressions, the coefficients on corruption exposure are significantly positive at the 5% level or better, suggesting that the higher the corruption exposure, the greater the degree of African operations.

We conduct a set of robustness checks for the positive relation between corruption exposure and African operation. First, we use different models for the tests. For example, we use a logit model instead of the probit model at the subsidiary-level. We use the Heckman selection model at the firm-level. Second, we use alternative measures of African operations. For example, we use the count of African subsidiaries instead of the percentage measures. Further, we use the dummy variable at the firm-level that equals one if the firm has operations in African, and otherwise zero. Finally, we use alternative control variables. For example, we use GDP per capita in the subsidiary-level tests and replace total assets by total capitalization in the firm-level tests. Throughout all of these tests, the positive relation between corruption and African operations consistently holds.

4.2 Corruption, African operations and stock returns

This section reports how stock returns are associated with corruption and African operations. For each firm, we identify its degree of corruption and whether it operates in Africa. The returns are all measured in U.S. dollars.

4.2.1 *Univariate test*

To begin with, we investigate how the annual stock return is related to African operations and corruption exposure. For a given year, we obtain the annual return by taking the geometric average of the stock's daily excess returns. [Table 4](#) presents return results for subsamples defined by corruption exposure and African operation status. Panel A of [Table 4](#) reports the results using *CPI* as the measure of corruption and Panel B of [Table 4](#) reports the results using *CPI* as the measure

of corruption. Overall, African operations with higher degrees of corruption are associated with significantly larger returns than non-African operations with lower degrees of corruption.

[\[Table 4\]](#)

While the stock returns for the firms with African operations in general is larger than for the firms without African operations, the results are only significant for firms with a higher degree of corruption exposure. In Panel A of [Table 4](#), where using *CPI* as the measure of corruption, the difference for firms with a higher degree of corruption exposure is 0.0240 (*t-statistics* = 3.42). In Panel B of [Table 4](#), where using *CCI* as the measure of corruption, the difference for firms with a higher degree of corruption exposure is 0.0247 (*t-statistics* = 3.49). Both results are significantly different from zero at the 1% level. For firms with a lower degree of corruption exposure, the difference between firms with and without African operations is 0.0011 in Panel A (*CPI* measure) and 0.0006 in Panel B (*CCI* measure). However, these results are not significant.

[Table 4](#) also documents that for firms with African operations, stock returns of firms with high corruption exposure are higher than those of firms with low corruption exposure. But for firms without African operations, stock returns of firms with high corruption exposure are lower than those of firms with low corruption exposure, though these results are not statistically significant. Importantly, we find that firms with high corruption exposure and meanwhile operating in Africa exhibit significantly higher returns than firms with low corruption exposure and without African operations. This is associated with a 0.0208 difference in annual return with the 1% significance level, using both *CPI* and *CCI* as measures of corruption. In fact, in four subsamples: firm-years with low corruption exposure and without African operations, firm-years with high corruption exposure and without African operations, firm-years with low corruption exposure and with

African operations, and firm-years with high corruption exposure and with African operations, only the subsample that firm-years with high corruption exposure and with African operations demonstrate positive returns.

Although these findings suggest the possibility of some interaction effects between corruption exposures and operations in Africa. However, they neither tell us whether these returns are abnormal nor control any risk factors that may affect the stock return. In the following section, we use the portfolio analysis to address these issues.

4.2.2 Portfolio analysis

In this section, we create corruption exposure weighted portfolios. We form the portfolios by weighting each firm by its relative value of a given corruption exposure index. Thus, firms with a higher value of corruption exposure index are given a larger weight in a portfolio. The portfolios are rebalanced once a year and the weight given to stock i in the portfolio is,

$$W_{i,t}^p = \frac{CEI_{i,t}^p}{\sum_{i=1}^N CEI_{i,t}^p} \quad (4)$$

where p equals the portfolio for a particular corruption exposure index (CEI_CPI or CEI_CCI) and $CEI_{i,t}^p$ is the corruption exposure index value for firm i in year t .

After forming the portfolios, we obtain a time series of daily return to each portfolio. We then regress the time series of portfolio returns in excess of the risk-free rate on the *CAPM* model (excess value of market returns, or *MKT*), the Fama and French's (1992, 1993) three-factor (*MKT*, *SMB*, and *HML*) model, the four-factor (*MKT*, *SMB*, *HML*, and *MOM*) model (i.e. the Fama and French's (1992, 1993) three factors plus the momentum factor), the Fama and French's (2015)

five-factor (*MKT*, *SMB*, *HML*, *RMW*, and *CMA*) model, and the six-factor (*MKT*, *SMB*, *HML*, *RMW*, *CMA*, and *MOM*) model (i.e. Fama and French's (2015) five factors plus the momentum factor), and report the intercepts in [Table 5](#).

[\[Table 5\]](#)

We find that without Africa operations, the corruption exposure portfolios earn significantly negative abnormal returns relative to the asset pricing models we consider. In contrast, with Africa operations, the corruption exposure portfolios earn positive abnormal returns and that the evidence of abnormal returns is robust to both *CEI_CPI* (reported in Panel A) and *CEI_CCI* (reported in Panel B). Across the five asset pricing models, the alphas from corruption exposure weighted portfolios are consistently 0.3 basis points (bps) for firms without Africa operations and range between 0.4 and 0.5 basis points (bps) for firms with Africa operations. In addition, all other pricing factors are statistically significant, only except for *MOM* in the four-factor model and *CMA* in the six-factor model for firms with African operations. Overall, these results suggest that while corruption negatively affects firm value, the effect could reverse if the firm has operations in Africa. These results, however, is just indicative as there could be systematic differences in various characteristics of the firms comprising the four subsamples. To the extent that these other characteristics are correlated with returns, it is important to consider these factors in a multivariate framework.

4.2.3 Regression analysis

In this section, we investigate how high corruption exposure is related to abnormal returns by performing regression analyses of annual abnormal returns on the lagged corruption exposure indexes and other market-, firm- and country-level characteristics. We calculate a firm's abnormal

return for a given year as the stock's annual return (geographically compounded daily returns) minus the return to risk factors from the six-factor (*MKT*, *SMB*, *HML*, *RMW*, *CMA*, and *MOM*) model over the same period. We then regress annual abnormal returns on corruption exposure and other potential determinants.

$$AR_{i,t} = \beta_0 + \beta_1 High_CEI_{i,t-1} + \beta_2 High_CEI \text{ with African Sub}_{i,t-1} + \beta_3 Firm_{i,t-1} + \beta_4 Country_{i,t-1} + \varepsilon_{i,t-1} \quad (5)$$

where $AR_{i,t}$ is the annual abnormal return of firm i at the end of year t , $High_CEI_{i,t-1}$ is a dummy variable if firm i has an above-median corruption exposure and otherwise zero, $High_CEI \text{ with African Sub}_{i,t-1}$ is a dummy variable if firm i has a high corruption exposure and is operating in Africa,²³ $Firm_{i,t-1}$ is a vector of firm-level variables and $Country_{i,t-1}$ is a vector of country-level variables, respectively,²⁴ and $\varepsilon_{i,t-1}$ is an error term. As shown in Section 4.1, not all firms have operations in Africa. Obviously, firms make decisions and choose whether to operate in Africa. This choice introduces a possible self-selection bias into our observed sample. To control for potential selection bias, we use a two-stage model. In the first stage, we estimate a probit regression of whether a firm has a subsidiary in Africa with considering corruption exposure and a range of determinants of African operation in the context of each model. The probit regression is estimated annually. From this first-stage regression, which identifies the likelihood of a firm participating in the African operation, we calculate the inverse Mills ratio (*MIR*) each year and include *MIR* in our return regressions at the second stage. This helps control the likelihood

²³ We introduce these variables because our focus is the effect of high corruption exposure with African operations. When we only use *CEI*, however, the estimation results are consistent, suggesting that the relation between corruption exposure and abnormal returns is associated with African operations. The results will be discussed in detail in the following sections.

²⁴ Appendix II provides detailed information about these explanatory variables.

of self-selection in the group of firms with African operations. At the second stage, we use Fama and MacBeth's (1973) procedure to estimate the equation (5). The t -statistics of the average of the time-series estimates from these year-by-year/cross-sectional regressions are adjusted for heteroscedasticity and autocorrelation using the Newey and West (1987) method.

4.2.3.1 *A classic model*

This section describes classic cross-sectional determinants and examines the extent to which these characteristics explain abnormal returns. Prior studies examine the extent to which cross-sectional stock returns in the United States and suggest that firm size (Fama & French, 1992; Keim, 1983; Perez-Quiros & Timmermann, 2000), book-to-market ratio (Fama & French, 1992, 1995; Griffin & Lemmon, 2002; Jiang, 2010; Loughran, 1997), trading volume (Ajinkya & Jain, 1989; Barclay, Litzenberger, & Warner, 1990; J. Y. Campbell, Grossman, & Wang, 1993; Chae, 2005; Chordia & Swaminathan, 2002), and the stock volatility (Baillie & DeGennaro, 1990; French, Schwert, & Stambaugh, 1987; Zhang, 2006) are important fundamental predictors. Following this earlier work, we measure firm size using the natural logarithm of total assets ($Ln(Assets)$), measure book-to-market equity using the natural logarithm of book-to-market ratio ($Ln(BM)$), measure trading volume using the number of shares traded in thousands (*Trading volume*), and measure stock volatility using stock price volatility (*Price volatility*). We include these variables along with the corruption exposure index in the first-stage probit model. After obtaining the *MIR* from the first stage, we include *MIR* with all other variables into the second-stage and estimate Fama and MacBeth's (1973) procedure.

[\[Table 6\]](#)

The first-stage probit results (unreported) suggest that corruption exposure, as measured by both *CEI_CPI* and *CEI_CCI*, and $\ln(TA)$ are significantly and positively associated with the possibility of having operations in Africa, whereas the association on $\ln(BM)$ is significantly negative. The coefficients on *Trading volume* and *Price volatility* are insignificant. The results from the second-stage Fama and MacBeth (1973) regressions reported in the first column of [Table 6](#), where Panel A reports the results with *CEI_CPI* as the measure of corruption exposure, and Panel B reports the results with *CEI_CCI* as the measure of corruption exposure. These results reveal that the coefficients on *High_CEI* and *High_CEI with African Sub* are both positive and quite significant after controlling for these classic determinants of stock returns. Apart from *High_CEI* and *High_CEI with African Sub*, all the other explanatory variables are significant. In line with prior studies, abnormal returns are negatively related to $\ln(TA)$ and are positively related to $\ln(BM)$, *Trading volume* and *Price volatility*. Based on *F-tests*, which is 9.35 for *CEI_CPI* regression and 9.45 for *CEI_CCI* regression, we conclude that the joint effect with taking accounting for corruption exposure and African operations significantly determines stock abnormal returns.

4.2.3.2 Firm-level characteristics

The value of a firm may be related to the characteristics and structures of the firm. Importantly, multinational operations have been found to be closely associated with knowledge transfers across countries through research and development (R&D) activities and intangible assets investments, which in turn affect the value of multinationals (Allen & Pantzalis, 1996; Gao & Chou, 2015; Gerybadze & Reger, 1999; L. Gu, 2016; Morck & Yeung, 1991). Further, not only international diversification but also business diversification significantly affects firm value (Berger & Ofek, 1995; Choi, Hiraki, & Landi, 2014; Denis et al., 2002; Lang & Stulz, 1994; Markides & Ittner, 1994). Moreover, the idea that cash flow uncertainty affects stock returns is examined by several

studies, such as Almeida et al. (2004), Irvine and Pontiff (2009), Jensen (1986), and Zhang (2006). Following this prior study as well as study that discuss corruption on firm value, such as Borisov et al. (2015), we include the logarithm of intangibles to assets ratio ($\ln(Intangible/Assets)$), of R&D to expenditure ratio ($\ln(R\&D/Expenditure)$), and of the number of business segments ($\ln(Diversification)$), and use three-year standard deviation of operating cash flow to capture the effect of cash flow volatility (*CF volatility*).

The first-stage probit results (unreported) indicate that corruption exposure, as measured by both *CEI_CPI* and *CEI_CCI*, remains significantly and positively associated with the possibility of having operations in Africa. Further, the coefficients on $\ln(Intangible/Assets)$ and $\ln(Diversification)$ are also significant and positive, but they are not significant on $\ln(R\&D/Expenditure)$ and *CF volatility*. The results from the second-stage Fama and MacBeth (1973) regressions reported in the second column of [Table 6](#), where Panel A reports the results with *CEI_CPI* as the measure of corruption exposure, and Panel B reports the results with *CEI_CCI* as the measure of corruption exposure. The results reveal that while still positive, the coefficient on *High_CEI* become insignificant after we control a range of firm-level characteristics. However, the coefficient on *High_CEI with African Sub* remains positive and significant at the 5% level. These results also suggest that abnormal return is significantly associated with R&D, intangible, operating cash flow and business diversifications. It is in particular worth noting that business diversification negatively affects firm value, these results are in line with Berger and Ofek (1995), Denis et al. (2002), and Lang and Stulz (1994). Again, the *F-test* successfully rejects the null hypothesis that all the coefficients including *High_CEI* and *High_CEI with African Sub* are jointly equal to zero.

4.2.3.3 Country-level factors

This section examines the extent to which the differences in abnormal returns can be explained by country-level variables suggested in the literature. As discussed in Siegel et al. (2011), the firm value associated with foreign investments can be partially determined by the egalitarianism distance. It has been well-documented that law, language, religion, and geographic distance can affect firm value (J. L. Callen & Fang, 2015; Demirgüç-Kunt & Maksimovic, 1998; El Ghouli, Guedhami, Ni, Pittman, & Saadi, 2012; Grinblatt & Keloharju, 2001). Further, studies suggest that firm values are higher for firms operating in a better institutional environment (David McLean, Zhang, & Zhao, 2012; Fama & French, 1998; Harford et al., 2017). To incorporate these effects, we follow the approach of Siegel et al. (2011) and use Schwartz's (2014) to measure *egalitarianism distance* between home and host countries, include *Different law family* (weighted average of subsidiaries in countries that do not adopt the common law system), *Common language* (English is the official language), *Common religion* (protestant is the dominant religion), *Geographic distance* (a thousand of kilometers of biggest cities of host countries from the United States), *Log of host country GDP*, *Corporate tax rate difference*, *Minority investor protection*, and *Unemployment benefits*. All these measures are weighted average where the weight is the percentage of subsidiaries in the given country to the total number of subsidiaries of the firm.

The first-stage probit results (unreported), which serve the purpose to correct self-selection bias for the effect on stock returns, show that the estimated coefficients on *Different law family*, *Common religion*, *Geographic distance*, and *Minority investor protection* are significant and positive. These results suggest that a firm's operation involves a higher proportion of countries with the different legal system, with protestant religion, with greater geographical distances, and with better minority investor protection would more likely to have operations in Africa. More

important for our purpose, the results from the second-stage the Fama and MacBeth (1973) regressions reported in the third column of Table 6 suggest that when these variables are included in our regressions, the estimated coefficient on *High_CEI with African Sub* remains significantly positive. However, the estimated coefficient on the *High_CEI* is insignificant. In addition, *Common language* is negatively related to abnormal returns, whereas *Common religion*, *Corporate tax rate difference*, and *Minority investor protection* are positively related to abnormal returns. The *F-test* fails to reject the null hypothesis that the coefficients of all the variables, including *High_CEI with African Sub* and *High_CEI* are jointly equal to zero at conventional significance levels.

4.2.3.4 *The degree of the foreign operation*

It would be of importance to ensure that our measure of corruption exposure is not a proxy of and/or influence by the degree of foreign operations. A higher degree of foreign operation may indicate a great chance to deal with foreign governments, which may likely to increase the source of corruption. However, it may be also a proxy for experience or the learning effect of a firm in foreign markets (Benito & Gripsrud, 1992; Zeng, Shenkar, Lee, & Song, 2013). This learning effect, however, is not necessarily only related to the corruption culture. Since there is no consensus on the measure of the degree of foreign operations, we use four alternative measures: The number of foreign to total subsidiaries (*FNTN*), foreign assets to total assets (*FATA*), foreign sales to total sales (*FSTS*), foreign operating income to total operating income (*FITI*), to ensure that our results are unbiased.

The first-stage probit results (unreported) indicate that all measures of the degree of foreign operations, only except for *FATA*, are positive and significant, suggesting that firms with a higher degree of foreign operations are more like to have African subsidiaries as well. The second-stage

Fama and MacBeth (1973) regressions, reported in last four columns of Table 6, show that only foreign sales to total sales (*FSTS*) are consistently and significantly positive at the 10% level in both Panel A and Panel B of Table 6. But important to us, it shows that the coefficient estimates on *High_CEI with African Sub* are all positive and significant. The results suggest that corruption exposure has a standalone and significant effect on abnormal returns, rather than a proxy of and/or influence by the degree of foreign operations.

4.2.4 Identification strategy

The previous regression analyses may lack power because firms are subject to the influence of factors and conditions in the United States, which may affect both firm foreign operation decision and stock returns. Thus endogeneity arisen from omitted variables can be an important issue that weakens the test power of our analyses. A convenient “natural reference” to address this concern is Chinese firms as they are also significantly involved in African operations (H. Campbell, 2008; Chen, Dollar, & Tang, 2016).²⁵ First, U.S. domestic factors and conditions do not affect Chinese firms’ African operation decisions, which are exogenous regarding our tests. Second, the degrees of corruption between China and Africa are proximate, compared to that between the United States and Africa.²⁶ Hence, the stock returns of Chinese firms that operate in Africa are less likely to be abnormal. Finally, as U.S. firms that are operating in Africa may have both high and low corruption exposures, the results after matching identify and reflect the effect of corruption exposure.

We use the propensity score match approach to address this endogeneity issue (Roberts & Whited, 2013). Because the disclosure of foreign operation of Chinese firms is significantly limited, even

²⁵ For the topic regarding Chinese operations in Africa, see also relevant research from the International Monetary Fund (IMF) (such as <https://www.imf.org/~media/Files/Publications/DP/2017/44711-afrdp.ashx>) and the World Bank, (such as <https://openknowledge.worldbank.org/handle/10986/21788>).

²⁶ As shown in [Appendix 1.A](#) and [Appendix 1.B](#), the average *CPI* and *CCI* of China over the sample period are 3.56 and -0.50, respectively.

for publicly listed Chinese firms, we reliably identify the firms that have African operations using public announcements on their agreements and contracts with African firms or governments. A list of these Chinese firms and their business description are given in [Appendix III](#). We then match them with U.S. firms that have African operations by the market (*Beta*, *trading volume*, and *stock price volatility*) and corporate characteristics (*firm size*, *the book to market equity ratio*, *intangibles to total assets*, *business diversification*, *tax ratio*²⁷ and *cash flow volatility*). The data used to construct these variables are obtained from Datastream.²⁸ For our purpose, we match firms with high and low corruption exposures, respectively.

[\[Table 7\]](#)

[Table 7](#) report the results. We employ four average-treatment-effect-on-the-treated (ATET) techniques to ensure unbiased results. These techniques include ATET nearest neighbor, ATET radius matching, ATET kernel matching, and ATET stratification matching. In Panel A, we report the results for all U.S. firms, which consist of firms with both high and low corruption exposure, after matching market and firm characteristics of Chinese firms by propensity score. The results from four techniques are quite consistent: The U.S. listed firms with African operations have a near to 10% higher abnormal returns than the Chinese listed firms with African operations. All results are significant at the 1% level. In Panel B, we separately look at the firms with high and low corruption exposure, which is based on *CPI*. The results show a distinctive pattern: U.S. firms with high corruption exposure tend to have a larger abnormal return than those with low corruption exposure for their African operations. What is more, the results are even insignificant for U.S.

²⁷ The ratio of tax to EBIT.

²⁸ Even though there is a data limitation to construct some variables for Chinese firms, data for constructing these variables are sufficiently available in Datastream.

firms with low corruption exposure when using ATET nearest neighbor and ATET stratification matching. Whereas results for firms with high corruption exposure are consistently significant and positive. In Panel C, we use *CCI* as corruption measure to look at abnormal returns of firms with high and low corruption exposure. Except for ATET nearest neighbor matching, which does not find the Chinese matching peers for U.S. firms, all other results are consistent with those in Panel B. Taken together, these results suggest that with matching Chinese firms with African operations, endogeneity does not appear to drive the positive effect of corruption exposure on stock returns of U.S. firms with African operations.

4.2.5 Robustness checks

The results in the previous section indicate that a firm's corruption exposure and operation in Africa significantly affect stock abnormal returns in various techniques and specifications that include a substantial number of variables that can conceivably explain the cross-sectional abnormal returns. In this section, we carry out robustness tests to address several concerns for if we are able to come up with plausible determinants of abnormal returns that subsume the effect of corruption exposure.

4.2.5.1 Operations in different African countries

The motivation of this study is based in part on the fact that corruption is generally high in Africa, compared to that in the United States. However, one may argue that even if there is high corruption in Africa in general, the degrees of corruption among African countries are different from one another. It is therefore important to perform a test that associates returns with different African countries. To test this, we focus on those sample firms that have subsidiaries in Africa and analyze the effect of corruption exposure with controlling for corruption distance between the United States related to each corresponding African country, in addition to all other control variables we

employed previously. We then estimate OLS specifications with clustered standard error on the country. We also include year fixed effects to control for the global economic shocks and the business cycle. The result of the test is reported in Panel A of [Table 8](#).

[\[Table 8\]](#)

We find that the annual abnormal returns are positively and statistically significantly associated with firms with high corruption exposures, with 0.334 (t-statistic = 3.50) and 0.242 (t-statistic = 3.23) for *CPI* and *CCI* as measures of corruption, respectively. The coefficients on corruption distance, however, are negatively related to the annual abnormal returns with -0.037 for *CPI* as the measures of corruption and -0.059 for *CCI* as the measure of corruption. Both are significant at the 10% level (t-statistic = -1.99 and -1.76, respectively). All control variables show a consistent sign with the previous results.²⁹ These results suggest that while corruption differences between the United States and African countries negatively affect firm value, the positive relation between corporate corruption exposure and stock return is not driven by cross-country differences.

4.2.5.2 *The alternative measure of corruption*

To further investigate the robustness of our results, we consider two alternative measures of corruption that are validated by Wilhelm (2002). These two indexes are the index of the overabundance of regulation or unnecessary restriction of business (*Excess Regulation*) activity and the index of black market activity (*Blank Market*). Both measures cover 85 major countries in the world and score range from one to five.³⁰ For these measures, because a higher score stands for a greater degree of corruption, we do need to perform any score transformation as did for *CPI*

²⁹ In the reported results, we use *FNTN* as the measure of the degree of foreign operations because *FNTN* offers more observations. The results are consistent using alternative measures of the degree of foreign operations.

³⁰ Refer to Wilhelm (2002) for a detailed discussion on these two measures.

and *CCI*. Similar to the previous tests, we re-estimate two-stage analyses. The first stage uses a probit model to test how exposures to *Excess Regulation* and *Blank Market* is related to the possibility of African operations. The results show significant and positive coefficients on these alternative measures of corruption, which are the same as our prior results. After obtaining the *IMR*, the second stage uses Fama and MacBeth (1973) regressions specified in equation (5) and investigate how corruption exposure with these two measures associated with African operations to affect abnormal returns. The results, reported in the Panel B of [Table 8](#), show that the estimated coefficients on *High CEI with African Sub* are positive (0.513 and 0.405 for *Excess Regulation* and *Blank Market*, respectively) and significant at the 1% level (t-statistic =3.07 and 2.92 for *Excess Regulation* and *Blank Market*, respectively). The results indicate that using alternative measures of corruption does not change our findings.

4.2.5.3 Does state-level corruption matter?

Some may argue that the effect of our measures of country-level exposures may be equal to state-level corruptions. This is a relevant issue since empirical studies also show that state-level corruption has a significant effect on financial activities and practices (Butler et al., 2009; Dass et al., 2016; Smith, 2016). If country-level corruption exposure is a proxy for state-level exposure, then two levels of corruption would produce identical results. To investigate this possibility, we replace country-level exposures by state-level exposures measured by *the average number of corruption-related convictions per million population* according to the number reported in Butler et al. (2009), Dass et al. (2016), and Smith (2016), respectively.³¹ In addition to this change, we further control state-level factors, including *population*, *median household income*, and *GDP* of

³¹ It is worth noting that Smith (2016) uses district-level data. We aggregate this data into the state-level according to the firm-year observations offered by the author.

each state.³² With these variables, we re-estimate two-stage analyses. The first stage uses a probit model to test how state-level corruption exposure is related to the possibility of African operations, and we obtain insignificant coefficients on all three measures of state-level corruption. In the second stage analysis, we investigate how state-level corruption exposure with African operations is associated abnormal returns by including *IMR* (obtained from the first stage) and using Fama and MacBeth (1973) regressions specified in equation (5) with standard error clustered on each state. The results in the Panel C of Table 8 show that the estimated coefficients on *High state corruption with African Sub* are negative (-0.513, -0.212 and -0.206 with state corruptions reported in Butler (2009), Dass (2016), and Smith (2016), respectively) and all statistically significant at the 5% level (t-statistic = -2.62, -2.82 and -2.70 with state corruptions reported in Butler (2009), Dass (2016), and Smith (2016), respectively). Hence, operating in Africa negatively affect the stock returns for firms from high corrupt states, suggesting that the positive relation between country-level corruption exposures and stock returns are different from and thus not a proxy for state-level corruption exposures.

4.2.5.4 Alternative control variables

To ensure that the results in the previous section are not driven by the quality of control variables, we consider alternative controls. While *total assets* is a well-accepted measure for firm size in the literature, it is also common to use *market capitalization* to capture the effect of firm size. We therefore replace the *natural logarithm of total assets* with the *natural logarithm of total market capitalization* in the previously reported regressions, finding very similar results. We also use other

³² We take natural logarithm for these state-level variables.

measures of firm size, such as *total capital* and *total sales*. Using these variables produces qualitatively similar results.

Although we controlled for egalitarianism distance that is documented by Siegel et al. (2011) as the most relevant cultural trait to affect cross-border financial activities, there is a large number of studies suggest that corruption may be related to other dimensions of national culture (See also, for instance, Husted (1999), Khatri, Tsang, & Begley (2006) Kwok & Tadesse (2006), Zheng, El Ghouli, Guedhami, & Kwok (2013). In particular, El Ghouli et al. (2016) document that corruption is negatively associated with individualistic culture. We thus use individualism as well as other cultural dimensions instead of egalitarianism to capture cultural distance. The results from these tests are qualitatively similar to our main results.

Because we do not find consistent and significant evidence in the portfolio analyses that the momentum effect is associated with African operations and corruption exposure to affect stock returns, we do not include the momentum factor in our regression analyses. However, it is interesting to test whether our results are affected by including the momentum. Unsurprisingly, we find a positive relation between the current and lagged returns, which is significant at the 1% level. Important to our study, including the lagged return does not change the positive and significant effect of high corruption exposures with African operations.

4.2.5.5 Further robustness

We perform a number of further robustness tests. To explore whether the effect of African operation and corruption exposure that we documented are driven by the global financial crisis, we estimate regressions of three subsample periods: 2004-2006, 2007-2009 and 2010-2014, the periods of before, during, and after the crisis. The results show that the positive effect of African

operation and corruption exposure are statistically significant before and after the financial crisis, but insignificant during the crisis period. This result is consistent with the assumption that the global financial crisis disrupts the market, suggesting that the positive effect of African operation and corruption exposure on stock returns are more likely to take place during the “normal” market time.

Further, we control for firm age. Newly listed firms may have extra cash on hand from their initial public offerings, and so may have more cash for corruption purpose and may be more likely to participate corrupt activities, which may result in a stronger effect of African operation and corruption exposure on stock returns. We re-estimate our regressions of Table 6 by including firm age from the year of initial public offerings as an independent variable. The coefficient on age is positive and significant, while the sign and significance level on the African operation and corruption exposure are relatively unchanged: the coefficient on the interaction of corruption exposure and African operation is 0.122 for *CEI_CPI* and 0.131 for *CEI_CCI*. Both coefficients are statistically significant at the 5% level.

Since smaller firms tend to generate higher returns, the positive relation between stock return and corruption exposure may be driven by smaller firms operating in African. To investigate this possibility, we remove all stock with a market capitalization less the 5th percentile (US\$36.7 million) from our sample. This forces us to reduce the number of observations (435 observations). Based on this reduced sample, we find that the abnormal returns (t-statistics), which is very close to the results reported in Table 6. We also examine a sample that excludes the largest stocks, since these stocks may be influenced by the trades of foreign institutions and thus may be less influenced by corruption cultures. Specifically, we remove stocks larger than the 95th percentile (US\$26.6 billion, 436 observations) from our sample. Similar findings are obtained. Hence, the positive

effect of the African operation and corruption exposure on stock returns is not driven by either small or big firms in our sample.

Finally, we adopt alternative measures of firm performance. If firms with higher degrees of corruption exposure benefit economically by participating in African operations, these benefits should be reflected in firm fundamentals, such as increases in profitability. In this final section, we estimate yearly regressions of changes in earnings return ($ROC(t) - ROC(t-1)$, where the return on capital (ROC) is equal to earnings before extraordinary items scaled by the book value of total capital). After controlling for potential sample selection bias in terms of which firms operating in Africa by including IMR from the first-stage probit model, we find a positive and significant effect of African operation and corruption exposure on the change in earning returns. This is consistent with our previous analyses where the stock return is the dependent variable. The result holds when we use ROA (return on assets) instead of ROC as the measure of earnings returns, as well as when we use level ROC instead of changes in ROC . The finding that corruption exposure has a standalone positive effect on earnings of multinational firms is in line with the literature (Henisz, 2000; Hines Jr, 1995; Huntington, 2006; Leff, 2002; Wei, 2000), suggesting corruption exposure contributes to firm profitability in the setting of international operations.

5 Conclusion

Corruption is a culture (DeBacker et al., 2015; Liu, 2016; Zingales, 2015), which may represent an unwritten rule that the real costs of public services in corrupt countries usually include tips of bribery. In these areas, local companies are accustomed to paying bribery for doing business. For foreign companies, however, they need to fit into the local environment in order to do business in these regions. The familiarity with corruption culture, measured by the country-level corruption

exposure, would enhance a firm's ability to adapt to the corrupt environment better and therefore gain benefits to shareholders. Using Africa as an experimental field, this paper finds that a higher corruption exposure contributes positively to the value of U.S. multinational firms.

Overall, the results of this paper suggest that it benefits to a firm if the level of corruption exposure of the firm matches the corruption culture of its destination markets. Future research is expected to extrapolate from the findings in this paper beyond the case of investing in Africa, particularly to identify channels through which corruption here proves beneficial.

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Table 1: The operations of U.S. listed firms in the world and in Africa

	Number of Subsidiaries	Percent (%)
Panel A: Operations in the world		
America	496,696	68.35
Europe	150,101	20.66
Asia	58,872	8.10
Pacific	13,359	1.84
Africa	7,635	1.05
Total	726,663	100.00
Panel B: Operations in Africa		
South Africa	2,773	36.32
Mauritius	1,123	14.71
Egypt	750	9.82
Nigeria	475	6.22
Morocco	348	4.56
Liberia	314	4.11
Kenya	258	3.38
Tunisia	199	2.61
Zimbabwe	178	2.33
Botswana	113	1.48
Ghana	108	1.41
Algeria	96	1.26
Tanzania	90	1.18
Namibia	75	0.98
Zambia	74	0.97
Uganda	73	0.96
Angola	66	0.86
Malawi	57	0.75
Cameroon	49	0.64
Equatorial Guinea	49	0.64
Senegal	48	0.63
Mozambique	40	0.52
Congo-Brazzaville (Republic)	29	0.38
Swaziland	22	0.29
Côte d'Ivoire	21	0.28
Gabon	20	0.26
Lesotho	19	0.25
Madagascar	18	0.24
Congo-Kinshasa (DR)	17	0.22
Mali	17	0.22
Guinea	15	0.20
Burkina Faso	14	0.18
Seychelles	14	0.18
Niger	13	0.17
Mauritania	10	0.13
Reunion	10	0.13
Chad	7	0.09
Ethiopia	7	0.09
Sierra Leone	7	0.09
Benin	5	0.07
Libya	5	0.07
Togo	3	0.04
Central African	2	0.03
Djibouti	1	0.01
Rwanda	1	0.01
Somalia	1	0.01
Sudan	1	0.01
Total	7,635	100.00

Table 2: Summary statistics

This table presents data of U.S. listed companies on New York stock exchange (NYSE), NASDAQ, and AMEX over a period from 2004 to 2014. All variables are estimated based on firm-year observations. *Mean return* is computed as the geometric mean of the daily return after subtracting risk-free rate. *Abnormal return* is calculated as the geometric compound the daily abnormal returns after matching a firm's stock return with Fama and French's (2015) five factors as well as momentum (the average return on the two high prior return portfolios minus the average return on the two low prior return portfolios). *CEI_CPI* is a corruption exposure index constructed according to Transparency International's Corruption Perceptions Index. *CEI_CCI* is a corruption exposure index constructed based on the World Bank's Control of Corruption Index. The data used to construct firm-level control variables, including abnormal returns are from DataStream, and the data used to construct country-level variables and for robustness checks are from various sources. The descriptions and sources of all variables are detailed in Appendix II.

Variable	N	Mean	Median	Std. Dev.	Min	Max
Mean return	8,853	0.77	0.66	3.27	-10.27	10.19
Abnormal return	8,848	0.21	0.02	1.55	-1.13	15.17
CEI_CPI	8,858	2.72	2.65	0.61	0.65	7.44
CEI_CCI	8,858	1.10	1.07	0.28	0.11	3.44
LnTA	8,720	13.78	13.71	1.91	6.50	20.50
LnBTM	8,452	-0.83	-0.78	0.77	-3.22	4.23
Share traded (in million)	8,858	2.13	0.49	7.92	0.00	311.43
Price volatility	8,189	35.65	34.40	12.38	11.37	81.13
Ln(Intangible/Assets)	7,796	2.53	2.87	1.37	-4.71	4.52
Ln(RD/Expense)	5,279	1.69	1.94	1.43	-5.10	4.29
Diversification	8,769	3.67	3.00	2.00	1.00	8.00
CF volatility	2,503	0.47	0.28	0.82	0.00	23.84
Egalitarianism distance	8,833	0.12	0.11	0.10	0.00	0.59
Difference in law (%)	8,833	22.18	14.29	24.55	0.00	100.00
Common language (%)	8,833	77.82	85.71	24.55	0.00	100.00
Common religion (%)	8,833	75.01	79.31	23.11	0.00	100.00
Geographic distance (mkm)	8,833	2.60	2.26	2.27	0.00	16.01
Log of host country GDP	8,832	29.74	29.90	0.59	25.71	30.33
Corporate tax rate	8,833	39.60	40.13	4.38	12.50	45.20
Minor investors protection	8,833	66.13	65.62	2.91	45.67	81.67
Underemployment benefits	8,833	0.30	0.27	0.13	0.15	1.00
FNTN (%)	8,858	46.20	47.06	29.58	0.00	100.00
FATA (%)	6,945	14.80	5.91	20.71	0.00	92.37
FSTS (%)	7,855	33.42	30.91	26.44	0.00	99.00
FITI (%)	6,161	27.60	15.40	72.08	-252.26	375.28
Excess regulation	8,858	2.27	2.19	0.33	1.00	4.00
Black market	8,858	1.23	1.04	0.37	1.00	5.00
Conviction Butler(2009)	2,949	2.95	2.73	1.10	0.73	4.95
Conviction Dass(2016)	2,949	2.98	2.60	1.04	0.93	5.68
Conviction Smith(2016)	2,949	3.18	2.56	1.53	0.89	7.49

Table 3: Corruption exposure and the possibility of operations in Africa

This table reports the regression of the occurrence of African operation on corruption exposures controlling for other relevant factors. Panel A conducts subsidiary-level analysis with probit model, where the dependent variable is a dummy variable that equals one if the subsidiary is in Africa and zero otherwise. Panel B conducts firm-level analysis with tobit model, where the dependent variable is the percentage of the firm's subsidiaries in Africa. All other independent variables are defined in Appendix II. The standard errors in Panel A are clustered on the country and in Panel B are clustered on the firm; the *t-statistics* are reported in the parentheses. ***, **, and * represent the significance level of 10%, 5%, and 1%, respectively.

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Subsidiary-level analysis								
Log of host country GDP	-0.552*** (-6.20)	-0.517*** (-4.38)	-0.530** (-2.39)	-0.146 (-0.30)	-0.553*** (-6.21)	-0.518*** (-4.40)	-0.532** (-2.40)	-0.147 (-0.31)
CEI_CPI	0.286*** (5.29)	0.147** (2.28)	0.209*** (4.96)	0.204** (2.57)				
CEI_CCI					0.628*** (5.05)	0.312** (2.20)	0.453*** (4.87)	0.444** (2.24)
Egalitarianism distance		-2.530*** (-4.12)	-6.781*** (-2.89)	-8.614** (-2.55)		-2.531*** (-4.12)	-6.782*** (-2.89)	-8.615** (-2.54)
Different law family			2.208* (1.86)	1.617 (0.90)			2.206 (1.42)	1.611 (0.80)
Common language			3.041*** (2.65)	2.869*** (3.34)			3.040*** (5.24)	2.861* (1.82)
Common religion			2.597** (2.44)	2.794*** (2.58)			2.599** (2.44)	2.798*** (2.58)
Geographic distance			-0.002 (-0.03)	-0.015 (-0.13)			-0.003 (-0.03)	-0.015 (-0.13)
Corporate tax rate difference				-0.091* (-1.71)				-0.091* (-1.71)
Minority investor protection				-0.116** (-2.02)				-0.116** (-2.01)
Unemployment benefits				0.279 (0.13)				0.263 (0.12)
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo- R^2	0.35	0.45	0.68	0.70	0.35	0.45	0.68	0.70
Observations	654,342	652,826	229,137	229,137	654,341	652,825	229,136	229,136

Continue to Table 3

Panel B: Firm-level analysis								
Log of total assets	0.433*** (16.04)	0.458*** (12.98)	0.489*** (11.22)	0.482*** (11.05)	0.431*** (16.07)	0.459*** (13.00)	0.490*** (11.23)	0.483*** (11.04)
CEI_CPI	0.793*** (13.06)	0.868*** (10.65)	1.155*** (9.53)	1.341*** (10.25)				
CEI_CCI					1.629*** (12.30)	1.806*** (10.08)	2.474*** (9.31)	2.887*** (10.07)
Intangible assets / Total assets		0.117** (2.45)	0.089 (1.63)	0.087 (1.54)		0.114** (2.40)	0.092* (1.67)	0.091 (1.60)
R&D / Operating expenditure		-0.055 (-0.97)	-0.127* (-1.87)	-0.126* (-1.84)		-0.05 (-0.89)	-0.125* (-1.84)	-0.124* (-1.81)
Business diversification		0.047 (0.43)	0.12 (1.02)	0.099 (0.84)		0.043 (0.40)	0.117 (1.00)	0.094 (0.80)
Egalitarianism distance			-5.709** (-2.47)	-7.113*** (-2.71)			-5.979** (-2.55)	-7.672*** (-2.90)
Different law family			0.530*** (3.39)	0.166 (0.95)			0.551*** (3.51)	0.202 (1.17)
Common language			0.785** (2.29)	0.756 (1.64)			0.730** (2.14)	0.608 (1.34)
Common religion			1.826*** (7.15)	2.102*** (7.83)			1.830*** (7.09)	2.131*** (7.89)
Geographic distance			0.402*** (6.39)	0.382*** (5.22)			0.395*** (6.20)	0.372*** (5.05)
Log of host country GDP			-1.245*** (-4.13)	-0.53 (-1.43)			-1.319*** (-4.29)	-0.607 (-1.64)
Corporate tax rate difference				-3.259** (-2.21)				-3.300** (-2.24)
Minority investor protection				-0.171 (-0.06)				0.569 (0.21)
Unemployment benefits				1.723*** (3.82)				1.695*** (3.71)
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo- R^2	0.36	0.41	0.48	0.49	0.35	0.40	0.48	0.49
Observations	11,751	6,474	4,866	4,866	11,755	6,474	4,866	4,866

Table 4: Univariate analysis

This table reports the results of univariate analysis for annual returns of U.S. listed firms with low and high corruption exposure, and with and without African operation. Firms with low and high corruption exposure are determined by the median of the degree of corruption exposure. Panel A measures corruption by Transparency International's Corruption Perceptions Index (*CPI*) and Panel B measures corruption by the World Banks' Control of Corruption Index (*CCI*). A firm has African operations if the firm has at least one subsidiaries in Africa. *Mean return* is computed as the geometric mean of the daily return after subtracting risk-free rate. The test for the difference of the means is measured using a standard two-tailed t-test, and the t-statistics are reported in the parentheses. ***, **, and * represent the significance level of 10%, 5%, and 1%, respectively.

	High corruption exposure (High)	Low corruption exposure (Low)	High minus Low (t-statistics)
Panel A: Using CPI as the measure of Corruption			
With African operation (With)	0.0031	-0.0162	0.0193 (1.21)
Without Africa operation (Without)	-0.0209	-0.0151	-0.0058 (1.17)
With minus Without (t-statistics)	0.0240*** (3.42)	0.0011 (0.03)	0.0208*** (3.12)
Panel B: Using CCI as the measure of Corruption			
With African operation (With)	0.0032	-0.0141	0.0173 (1.15)
Without Africa operation (Without)	-0.0215	-0.0147	-0.0068 (1.40)
With minus Without (t-statistics)	0.0247*** (3.49)	0.0006 (0.03)	0.0208*** (3.10)

Table 5: Abnormal return for firms according to corruption portfolio

This table reports the portfolio analysis of daily abnormal returns for firms *without* and *with* operations in Africa. A firm has African operations if the firm has at least one subsidiaries in Africa. Portfolios are formed by weighting each firm by its relative value of a given lagged corruption exposure. Panel A measures corruption by Transparency International's Corruption Perceptions Index (*CPI*) and Panel B measures corruption by the World Banks' Control of Corruption Index (*CCI*). The portfolio is rebalanced once a year. The weight given to stock *i* in the portfolio is,

$$W_{i,t}^p = \frac{CEI_{i,t}^p}{\sum_{i=1}^N CEI_{i,t}^p}$$

where *p* equals the portfolio for a particular corruption exposure index (*CEI_CPI* or *CEI_CCI*) and $CEI_{i,t}^p$ is the corruption exposure index value for firm *i* in year *t*. We form a time series of daily returns to each portfolio from 2004 to 2014. We regress the time series of portfolio returns in the excess of the risk-free rate on the market risk premium from the market model (*CAPM*), Fama and French (1992, 1993) model (*Three-factor*), Fama and French's (1992, 1993) model plus Carhart's (1997) momentum factor (*Four-factor*), Fama and French's (2015) five-factor model (*Five-factor*), and Fama and French's (2015) five-factor model plus Carhart's (1997) momentum factor (*Six-factor*) for each portfolio. Returns are in decimal form, that is, 0.01 is 1%. *T*-statistics are in parentheses. ***, **, and * represent the significance level of 10%, 5%, and 1%, respectively.

Model	Africa operation	Alpha	MKT	SMB	HML	MOM	RMW	CMA
Panel A: CEI_CPI weighted portfolios								
CAPM	Without	-0.00003*** (-16.45)	0.00072*** (716.99)					
	With	0.00005 (0.92)	0.00115*** (34.62)					
Three-factor	Without	-0.00003*** (-19.23)	0.00068*** (584.40)	0.00054*** (225.59)	0.00005*** (22.71)			
	With	0.00004 (0.84)	0.00119*** (30.68)	0.00027*** (3.42)	-0.00024*** (-3.34)			
Four-factor	Without	-0.00003*** (-19.24)	0.00068*** (567.38)	0.00054*** (224.46)	0.00005*** (18.72)	0.00000 (-1.51)		
	With	0.00004 (0.85)	0.00121*** (30.25)	0.00026*** (3.23)	-0.00018** (-2.10)	0.00008 (1.51)		
Five-factor	Without	-0.00003*** (-16.66)	0.00066*** (551.82)	0.00053*** (211.79)	-0.00010*** (-40.27)		-0.00021*** (-43.61)	0.00004*** (7.71)
	With	0.00005 (0.96)	0.00119*** (29.36)	0.00021** (2.53)	-0.00038*** (-4.66)		-0.00032** (-1.99)	0.00024 (1.42)
Six factor	Without	-0.00003*** (-16.59)	0.00066*** (543.55)	0.00053*** (210.67)	-0.00009*** (-33.24)	0.00001*** (4.18)	-0.00021*** (-43.80)	0.00003*** (6.63)
	With	0.00005 (0.99)	0.00120*** (29.20)	0.00020** (2.41)	-0.00032*** (-3.37)	0.00008 (1.40)	-0.00033** (-2.07)	0.00018 (1.05)
Panel B: CEI_CCI weighted portfolios								
CAPM	Without	-0.00003*** (-16.28)	0.00072*** (710.75)					
	With	0.00005 (0.91)	0.00116*** (34.98)					
Three-factor	Without	-0.00003*** (-19.03)	0.00067*** (579.00)	0.00053*** (223.13)	0.00005*** (23.03)			
	With	0.00004 (0.83)	0.00121*** (30.98)	0.00028*** (3.47)	-0.00024*** (-3.34)			
Four-factor	Without	-0.00003*** (-19.03)	0.00067*** (562.18)	0.00053*** (221.99)	0.00005*** (19.10)	0.00000 (-1.29)		
	With	0.00004 (0.84)	0.00122*** (30.54)	0.00026*** (3.28)	-0.00018** (-2.10)	0.00008 (1.52)		
Five-factor	Without	-0.00003*** (-16.52)	0.00066*** (546.81)	0.00052*** (209.58)	-0.00009*** (-39.03)		-0.00020*** (-42.51)	0.00004*** (7.59)
	With	0.00005 (0.95)	0.00120*** (29.65)	0.00022*** (2.59)	-0.00038*** (-4.67)		-0.00032** (-1.98)	0.00024 (1.41)
Six factor	Without	-0.00003*** (-16.46)	0.00066*** (538.64)	0.00052*** (208.46)	-0.00009*** (-32.09)	0.00001*** (4.31)	-0.00020*** (-42.72)	0.00003*** (6.49)
	With	0.00005 (0.98)	0.00121*** (29.48)	0.00021** (2.46)	-0.00032*** (-3.37)	0.00008 (1.41)	-0.00033** (-2.06)	0.00018 (1.03)

Table 6: Determinants of abnormal return: Results from the two-stage panel regressions for African operations and corruption profits

This table presents regressions of annual abnormal returns, calculated from Fama and French's (2015) five-factor model plus Carhart's (1997) momentum factor, on lagged corruption exposure, control variables, and the inverse Mills ratio (IMR). *High_CEI* is the degree of corruption exposure above the median. Panel A measures corruption by Transparency International's Corruption Perceptions Index (CPI). Panel B measures corruption by the World Banks' Control of Corruption Index (CCI). *High_CEI with African Sub* is firms that have African operations and with high corruption exposure. We control for market related variables (*Ln(Assets)*, *Ln(BM)*, *Trading volume*, and *Price volatility*), firm-level variables (*Ln(Intangible/Assets)*, *Ln(R&D/Expenditure)*, *Ln(Diversification)*, and *CF volatility*), country-level variables (*Egalitarianism distance*, *Different law family*, *Common language*, *Common religion*, *Geographic distance*, *Log of host country GDP*, *Corporate tax rate difference*, *Minority investor protection*, and *Unemployment benefits*), and four different measure of the degree of foreign operations (*FNTN*, *FATA*, *FSTS*, *FITI*). Details on the description and construction of these variables are provided in Appendix II. *T*-statistics based on White-corrected robust standard errors clustered on the firm are in parentheses. ***, **, and * represent the significance level of 10%, 5%, and 1%, respectively.

Model	Classic	Firm	Country	Foreign operation			
Panel A: Using CPI as the measure of Corruption							
Intercept	24.306*** (3.95)	1.814*** (6.86)	-37.715* (-2.26)	-35.973* (-2.19)	-53.212** (-3.07)	-34.470* (-2.19)	-32.400* (-2.06)
IMR	-3.747** (-3.37)	-0.521** (-2.72)	-0.339** (-2.78)	-0.365** (-2.95)	-0.330** (-2.65)	-0.322** (-2.78)	-0.396*** (-4.15)
<i>High_CEI</i>	2.967** (2.45)	0.012 (0.20)	-0.054 (-0.71)	-0.077 (-0.95)	-0.004 (-0.03)	-0.037 (-0.49)	-0.095 (-1.06)
<i>High_CEI with African Sub</i>	6.091** (3.15)	1.049** (2.55)	0.812** (2.58)	0.821** (2.74)	0.800** (2.51)	0.781** (2.55)	0.897** (2.77)
Ln(Assets)	-2.198*** (-3.80)	-0.095*** (-4.61)	-0.044 (-1.34)	-0.048 (-1.48)	-0.03 (-0.90)	-0.034 (-0.91)	-0.055 (-1.58)
Ln(BM)	5.492** (2.55)	0.612** (2.74)	0.458** (2.49)	0.452** (2.49)	0.561** (2.74)	0.435* (2.43)	0.524** (2.64)
Trading volume	0.111** (2.61)	-0.006 (-1.27)	-0.005 (-1.07)	-0.005 (-0.95)	-0.004 (-1.01)	-0.007 (-1.23)	-0.003 (-1.07)
Price volatility	0.406** (3.23)	0.094*** (5.48)	0.088*** (5.52)	0.088*** (5.47)	0.085*** (5.64)	0.090*** (5.36)	0.088*** (5.48)
Ln(Intangible/Assets)		-0.197*** (-3.88)	-0.166** (-3.39)	-0.164** (-3.45)	-0.086*** (-4.19)	-0.156** (-3.30)	-0.199*** (-4.39)
Ln(R&D/Expenditure)		-0.221*** (-4.92)	-0.185*** (-5.39)	-0.188*** (-5.40)	-0.188*** (-5.11)	-0.197*** (-4.94)	-0.200*** (-4.59)
Ln(Diversification)		-0.115** (-2.90)	-0.144** (-3.12)	-0.164** (-3.35)	-0.114** (-2.55)	-0.112** (-2.98)	-0.150** (-2.59)
CF volatility		0.262** (3.52)	0.166* (2.36)	0.165* (2.35)	0.150* (2.08)	0.172** (2.59)	0.174** (2.62)
Egalitarianism distance			2.762 (1.23)	2.97 (1.21)	2.663 (0.94)	2.44 (0.99)	3.924 (1.35)
Different law family			-0.298 (-1.59)	-0.318 (-1.65)	-0.447 (-1.81)	-0.302 (-1.54)	-0.335** (-2.57)
Common language			-1.668** (-3.25)	-1.619** (-3.25)	-2.150** (-3.50)	-1.471** (-3.50)	-1.513** (-3.31)
Common religion			0.897** (2.63)	0.839** (2.57)	0.698** (2.68)	0.898** (2.75)	0.874** (3.17)
Geographic distance			0.037 (1.23)	0.028 (0.99)	0.051** (3.27)	0.035 (1.23)	0.042 (1.52)
Log of host country GDP			0.487 (1.15)	0.514 (1.19)	0.619 (1.48)	0.379 (0.90)	0.429 (1.46)
Corporate tax rate difference			1.199*** (3.81)	1.285*** (4.12)	1.026** (2.91)	1.499*** (3.75)	1.332 (1.63)
Minority investor protection			4.414** (2.92)	3.724** (2.60)	7.074*** (3.93)	4.097** (3.12)	3.481 (1.63)
Unemployment benefits			-0.171 (-0.52)	-0.176 (-0.55)	-0.332 (-1.41)	-0.035 (-0.10)	-0.175 (-0.61)
FNTN				0.108 (0.70)			
FATA					0.002 (1.83)		
FSTS						0.004* (2.41)	
FITI							0.00 (-1.19)
<i>F</i> -value	9.35	12.22	7.90	7.74	8.32	7.50	7.36
Avg. <i>R</i> ²	0.191	0.513	0.561	0.561	0.579	0.565	0.572
Number of observations	7,907	1,269	1,128	1,128	993	1,110	928

Continue to Table 6

Panel B: Using CCI as the measure of Corruption							
Intercept	24.513*** (3.98)	1.849*** (6.24)	-36.505* (-2.16)	-33.977* (-2.07)	-52.059** (-2.95)	-33.574* (-2.12)	-29.284* (-1.98)
IMR	-3.883** (-3.29)	-0.501** (-3.03)	-0.323*** (-3.77)	-0.347*** (-3.88)	-0.309** (-3.59)	-0.306** (-3.68)	-0.396*** (-4.15)
<i>High_CEI</i>	2.302** (2.45)	0.019 (0.36)	-0.037 (-0.57)	-0.054 (-0.75)	0.003 (0.03)	-0.026 (-0.40)	-0.066 (-1.00)
<i>High_CEI with African Sub</i>	6.640** (3.28)	1.037** (2.85)	0.813** (3.12)	0.820** (3.32)	0.793** (3.07)	0.782** (3.05)	0.901** (3.30)
Ln(Assets)	-2.197*** (-3.79)	-0.096*** (-4.55)	-0.049 (-1.35)	-0.052 (-1.44)	-0.035 (-0.97)	-0.039 (-0.95)	-0.066 (-1.75)
Ln(BM)	5.476** (2.55)	0.612** (2.80)	0.460** (2.54)	0.454** (2.54)	0.558** (2.81)	0.437** (2.49)	0.527** (2.70)
Trading volume	0.111** (2.61)	-0.006 (-1.23)	-0.005 (-0.93)	-0.004 (-0.80)	-0.003 (-0.84)	-0.006 (-1.13)	-0.003 (-0.78)
Price volatility	0.408** (3.24)	0.094*** (5.42)	0.088*** (5.43)	0.088*** (5.37)	0.085*** (5.49)	0.090*** (5.30)	0.088*** (5.35)
Ln(Intangible/Assets)		-0.202*** (-3.94)	-0.170** (-3.55)	-0.168** (-3.60)	-0.088*** (-4.71)	-0.159** (-3.47)	-0.209*** (-4.44)
Ln(R&D/Expenditure)		-0.222*** (-4.82)	-0.187*** (-5.04)	-0.189*** (-5.05)	-0.190*** (-4.85)	-0.198*** (-4.60)	-0.204*** (-4.36)
Ln(Diversification)		-0.113** (-2.99)	-0.143** (-3.17)	-0.163** (-3.49)	-0.113** (-2.85)	-0.111** (-3.07)	-0.150** (-2.62)
CF volatility		0.263** (3.67)	0.169* (2.43)	0.168* (2.43)	0.153* (2.16)	0.175** (2.67)	0.176** (2.73)
Egalitarianism distance			3.277 (1.41)	3.603 (1.40)	3.336 (1.12)	2.983 (1.17)	4.685 (1.57)
Different law family			-0.305 (-1.69)	-0.323 (-1.71)	-0.452 (-1.89)	-0.308 (-1.62)	-0.343** (-2.88)
Common language			-1.619** (-3.21)	-1.549** (-3.18)	-2.087** (-3.45)	-1.422** (-3.47)	-1.418** (-3.31)
Common religion			0.929** (2.64)	0.875** (2.60)	0.740** (2.67)	0.925** (2.75)	0.915** (3.14)
Geographic distance			0.029 (0.89)	0.02 (0.67)	0.046** (2.48)	0.03 (0.97)	0.027 (0.97)
Log of host country GDP			0.461 (1.09)	0.475 (1.10)	0.596 (1.39)	0.359 (0.85)	0.389 (1.35)
Corporate tax rate difference			1.340*** (3.89)	1.420*** (4.06)	1.214** (3.20)	1.656*** (3.77)	1.422 (1.70)
Minority investor protection			4.203** (2.99)	3.422** (2.66)	6.805*** (4.07)	3.902** (3.32)	2.987 (1.58)
Unemployment benefits			-0.159 (-0.48)	-0.165 (-0.50)	-0.327 (-1.34)	-0.028 (-0.08)	-0.162 (-0.56)
FNTN				0.072 (0.36)			
FATA					0.002* (1.96)		
FSTS						0.004* (2.19)	
FITI							0.00 (-1.31)
<i>F-value</i>	9.45	12.55	8.27	8.02	8.80	7.83	7.79
<i>Avg. R²</i>	0.191	0.513	0.56	0.562	0.579	0.564	0.572
Number of observations	7,907	1,269	1,128	1,128	993	1,110	928

Table 7: Propensity score match of African operations

This table reports the results of propensity score matches for African operations of U.S. listed firms with African operations of Chinese listed firms. Panel A reports the results for all U.S. firms. Panels B and C report the results for firms with high and low corruption exposures, respectively. In Panel B, corruption is measured by Transparency International's Corruption Perceptions Index (CPI), and in Panel C, corruption is measured by the World Bank's Control of Corruption Index (CCI). We match the U.S. and Chinese listed firms that have African operations with the market (Beta, trading volume, and stock price volatility) and corporate characteristics (firm size, the book to market equity ratio, the tax to EBIT ratio, intangibles to total assets, business diversification, and cash flow volatility). Details on the description and construction of these variables are provided in Appendix II. We use four different propensity score matching techniques: ATET nearest neighbor is the average effect of the treatment on the treated based on the nearest neighbor matching. ATET radius matching is the average effect of the treatment on the treated based on radius matching technique. ATET Kernel Match is the average effect of the treatment on the treated by matching with kernel weighting. ATET stratification matching is the average effect of the treatment on the treated by matching with stratification technique. The sample covers listed U.S. multinationals from 2004 to 2014. The results are obtained from 5 times bootstrapping. Numbers in parentheses report t-statistics based on White-corrected robust standard errors clustered on the firm. *, **, and *** stand for the 10%, 5%, and 1% significance levels, respectively.

	Dependent variable: Abnormal Return				
	Panel A	Panel B: CPI measuring corruption		Panel C: CCI measuring corruption	
	All U.S. firm with African operation	High CEI U.S. firm with African operation	Low CEI U.S. firm with African operation	High CEI U.S. firm with African operation	Low CEI U.S. firm with African operation
ATET nearest neighbor	9.930*** (15.17)	34.580*** (12.82)	0.805 (0.31)	7.033* (1.72)	-- --
ATET radius matching	9.951*** (18.86)	39.115*** (2.59)	4.316*** (2.59)	32.236*** (2.78)	5.789*** (2.90)
ATET kernel matching	9.945*** (10.98)	4.423*** (4.33)	1.901*** (10.29)	4.451*** (3.23)	1.959*** (17.84)
ATET stratification matching	9.532*** (10.06)	4.425*** (4.80)	1.915 (1.09)	4.450*** (3.54)	1.958** (2.24)

Table 8: Operations in African countries, state corruption, and alternative variables

This table presents regressions of annual abnormal returns, calculated from Fama and French's (2015) five-factor model plus Carhart's (1997) momentum factor, on lagged corruption exposure, which is measured by *CPI* and *CCI* in Panel A, by Wilhelm's (2002) *Excess Regulation* and *Blank Market* in Panel B, and by state-level corruption exposures of according to the number reported in Butler et al. (2009), Dass et al. (2016), and Smith (2016), respectively, in Panel C. *T*-statistics based on White-corrected robust standard errors clustered on the firm are in parentheses. ***, **, and * represent the significance level of 10%, 5%, and 1%, respectively.

Dependent variable	Abnormal returns		
	Panel A: Operations in African countries		
	CPI measuring corruption	CCI measuring corruption	
Model	(1)	(2)	
Intercept	-53.733*** (-8.91)	-55.669*** (-8.82)	
<i>High corruption exposure (CEI)</i>	<i>0.334***</i> (3.50)	<i>0.242***</i> (3.23)	
<i>Corruption distance</i>	<i>-0.037*</i> (-1.99)	<i>-0.059*</i> (-1.76)	
Controls:	Yes	Yes	
Year dummy	Yes	Yes	
Adjusted. <i>R</i> ²	0.491	0.488	
Number of observations	1,478	1,478	
	Panel B: The alternative measures of corruption		
	Excess Regulation	Blank Market	
Model	(1)	(2)	
Intercept	-29.293 (-1.90)	-29.148 (-1.93)	
IMR	-0.229*** (-3.88)	-0.158*** (-4.05)	
<i>High corruption exposure</i>	<i>-0.096</i> (-1.84)	<i>-0.076</i> (-1.43)	
<i>High corruption exposure with African Sub</i>	<i>0.513***</i> (3.07)	<i>0.405***</i> (2.92)	
State-level controls	Yes	Yes	
Firm and country level controls	Yes	Yes	
Avg. <i>R</i> ²	0.559	0.558	
Number of observations	1,128	1,128	
	Panel C: State-level corruption		
	Butler (2009)	Dass (2016)	Smith (2016)
Model	(1)	(2)	(3)
Intercept	-36.446 (-1.89)	-38.602 (-1.94)	-38.508 (-1.94)
IMR	0.158* (2.42)	0.173** (2.70)	0.169** (2.72)
<i>High State corruption</i>	<i>-0.077</i> (-1.23)	<i>-0.163</i> (-1.83)	<i>-0.167</i> (-1.84)
<i>High state corruption with African Sub</i>	<i>-0.206**</i> (-2.62)	<i>-0.212**</i> (-2.82)	<i>-0.206**</i> (-2.70)
State-level controls	Yes	Yes	Yes
Firm and country level controls	Yes	Yes	Yes
Avg. <i>R</i> ²	0.573	0.576	0.576
Number of observations	1,120	1,120	1,120

Appendix I.A: The Corruption Perception Index (CPI), 2004 - 2014

Country/Territory	Area	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	<i>CPI</i>
Botswana	AF	6.00	5.90	5.60	5.40	5.80	5.60	5.80	6.10	6.50	6.40	6.30	5.95
Cape Verde	AF				4.90	5.10	5.10	5.10	5.50	6.00	5.80	5.70	5.40
Mauritius	AF	4.10	4.20	5.10	4.70	5.50	5.40	5.40	5.10	5.70	5.20	5.40	5.07
Seychelles	AF	4.40	4.00	3.60	4.50	4.80	4.80	4.80	4.80	5.20	5.40	5.50	4.71
South Africa	AF	4.60	4.50	4.60	5.10	4.90	4.70	4.50	4.10	4.30	4.20	4.40	4.54
Namibia	AF	4.10	4.30	4.10	4.50	4.50	4.50	4.40	4.40	4.80	4.80	4.90	4.48
Ghana	AF	3.60	3.50	3.30	3.70	3.90	3.90	4.10	3.90	4.50	4.60	4.80	3.98
Rwanda	AF		3.10	2.50	2.80	3.00	3.30	4.00	5.00	5.30	5.30	4.90	3.92
Lesotho	AF		3.40	3.20	3.30	3.20	3.30	3.50	3.50	4.50	4.90	4.90	3.77
Burkina Faso	AF		3.40	3.20	2.90	3.50	3.60	3.10	3.00	3.80	3.80	3.80	3.41
Senegal	AF	3.00	3.20	3.30	3.60	3.40	3.00	2.90	2.90	3.60	4.10	4.30	3.39
Swaziland	AF		2.70	2.50	3.30	3.60	3.60	3.20	3.10	3.70	3.90	4.30	3.39
Sao Tome and Principe	AF				2.70	2.70	2.80	3.00	3.00	4.20	4.20	4.20	3.35
Djibouti	AF				2.90	3.00	2.80	3.20	3.00	3.60	3.60	3.40	3.19
Gabon	AF	3.30	2.90	3.00	3.30	3.10	2.90	2.80	3.00	3.50	3.40	3.70	3.17
Benin	AF	3.20	2.90	2.50	2.70	3.10	2.90	2.80	3.00	3.60	3.60	3.90	3.11
Malawi	AF	2.80	2.80	2.70	2.70	2.80	3.30	3.40	3.00	3.70	3.70	3.30	3.11
Liberia	AF		2.20		2.10	2.40	3.10	3.30	3.20	4.10	3.80	3.70	3.10
Zambia	AF	2.60	2.60	2.60	2.60	2.80	3.00	3.00	3.20	3.70	3.80	3.80	3.06
Madagascar	AF	3.10	2.80	3.10	3.20	3.40	3.00	2.60	3.00	3.20	2.80	2.80	3.00
Tanzania	AF	2.80	2.90	2.90	3.20	3.00	2.60	2.70	3.00	3.50	3.30	3.10	3.00
Mali	AF	3.20	2.90	2.80	2.70	3.10	2.80	2.70	2.80	3.40	2.80	3.20	2.95
Gambia	AF	2.80	2.70	2.50	2.30	1.90	2.90	3.20	3.50	3.40	2.80	2.90	2.81
Mozambique	AF	2.80	2.80	2.80	2.80	2.60	2.50	2.70	2.70	3.10	3.00	3.10	2.81
Niger	AF	2.20	2.40	2.30	2.60	2.80	2.90	2.60	2.50	3.30	3.40	3.50	2.77
Mauritania	AF			3.10	2.60	2.80	2.50	2.30	2.40	3.10	3.00	3.00	2.76
Ethiopia	AF	2.30	2.20	2.40	2.40	2.60	2.70	2.70	2.70	3.30	3.30	3.30	2.72
Togo	AF			2.40	2.30	2.70	2.80	2.40	2.40	3.00	2.90	2.90	2.64
Uganda	AF	2.60	2.50	2.70	2.80	2.60	2.50	2.50	2.40	2.90	2.60	2.60	2.61
Comoros	AF				2.60	2.50	2.30	2.10	2.40	2.80	2.80	2.60	2.51
Eritrea	AF	2.60	2.60	2.90	2.80	2.60	2.60	2.60	2.50	2.50	2.00	1.80	2.50
Sierra Leone	AF	2.30	2.40	2.20	2.10	1.90	2.20	2.40	2.50	3.10	3.00	3.10	2.47
Cameroon	AF	2.10	2.20	2.30	2.40	2.30	2.20	2.20	2.50	2.60	2.50	2.70	2.36
Nigeria	AF	1.60	1.90	2.20	2.20	2.70	2.50	2.40	2.40	2.70	2.50	2.70	2.35
Côte d'Ivoire	AF	2.00	1.90	2.10	2.10	2.00	2.10	2.20	2.20	2.90	2.70	3.20	2.31
Kenya	AF	2.10	2.10	2.20	2.10	2.10	2.20	2.10	2.20	2.70	2.70	2.50	2.27
Central African Republic	AF			2.40	2.00	2.00	2.00	2.10	2.20	2.60	2.50	2.40	2.24
Zimbabwe	AF	2.30	2.60	2.40	2.10	1.80	2.20	2.40	2.20	2.00	2.10	2.10	2.20
Congo Republic	AF	2.30	2.30	2.20	2.10	1.90	1.90	2.10	2.20	2.60	2.20	2.30	2.19
Guinea-Bissau	AF				2.20	1.90	1.90	2.10	2.20	2.50	1.90	1.90	2.08
Guinea	AF			1.90	1.90	1.60	1.80	2.00	2.10	2.40	2.40	2.50	2.07
Burundi	AF		2.30	2.40	2.50	1.90	1.80	1.80	1.90	1.90	2.10	2.00	2.06
Angola	AF	2.00	2.00	2.20	2.20	1.90	1.90	1.90	2.00	2.20	2.30	1.90	2.05
Democratic Republic of the Congo	AF	2.00	2.10	2.00	1.90	1.70	1.90	2.00	2.00	2.10	2.20	2.20	2.01
Equatorial Guinea	AF		1.90	2.10	1.90	1.70	1.80	1.90	1.90	2.00	1.90		1.90
Chad	AF	1.70	1.70	2.00	1.80	1.60	1.60	1.70	2.00	1.90	1.90	2.20	1.83
South Sudan	AF										1.40	1.50	1.45
Somalia	AF		2.10		1.40	1.00	1.10	1.10	1.00	0.80	0.80	0.80	1.12
Canada	AM	8.50	8.40	8.50	8.70	8.70	8.70	8.90	8.70	8.40	8.10	8.10	8.52
United States	AM	7.50	7.60	7.30	7.20		7.50	7.10	7.10	7.30	7.30	7.40	7.33
Barbados	AM	7.30	6.90	6.70	6.90	7.00	7.40	7.80	7.80	7.60	7.50	7.40	7.30
Bahamas	AM								7.30	7.10	7.10	7.10	7.15
Chile	AM	7.40	7.30	7.30	7.00	6.90	6.70	7.20	7.20	7.20	7.10	7.30	7.15
Saint Lucia	AM				6.80	7.10	7.00		7.00	7.10	7.10	7.10	7.03
Uruguay	AM	6.20	5.90	6.40	6.70	6.90	6.70	6.90	7.00	7.20	7.30	7.30	6.77
Saint Vincent and the Grenadines	AM				6.10	6.50	6.40		5.80	6.20	6.20	6.20	6.20
Puerto Rico	AM					5.80	5.80	5.80	5.60	6.30	6.20	6.30	5.97
Dominica	AM			4.50	5.60	6.00	5.90	5.20	5.20	5.80	5.80	5.80	5.53
Costa Rica	AM	4.90	4.20	4.10	5.00	5.10	5.30	5.30	4.80	5.40	5.30	5.40	4.98
Cuba	AM	3.70	3.80	3.50	4.20	4.30	4.40	3.70	4.20	4.80	4.60	4.60	4.16
El Salvador	AM	4.20	4.20	4.00	4.00	3.90	3.40	3.60	3.40	3.80	3.80	3.90	3.84
Brazil	AM	3.90	3.70	3.30	3.50	3.50	3.70	3.70	3.80	4.30	4.20	4.30	3.81
Colombia	AM	3.80	4.00	3.90	3.80	3.80	3.70	3.50	3.40	3.60	3.60	3.70	3.71
Trinidad and Tobago	AM	4.20	3.80	3.20	3.40	3.60	3.60	3.60	3.20	3.90	3.80	3.80	3.65
Peru	AM	3.50	3.50	3.30	3.50	3.60	3.70	3.50	3.40	3.80	3.80	3.80	3.58
Suriname	AM	4.30	3.20	3.00	3.50	3.60	3.70		3.00	3.70	3.60	3.60	3.52
Panama	AM	3.70	3.50	3.10	3.20	3.40	3.40	3.60	3.30	3.80	3.50	3.70	3.47
Jamaica	AM	3.30	3.60	3.70	3.30	3.10	3.00	3.30	3.30	3.80	3.80	3.80	3.45
Grenada	AM			3.50	3.40								3.45
Mexico	AM	3.60	3.50	3.30	3.50	3.60	3.30	3.10	3.00	3.40	3.40	3.50	3.38
Belize	AM	3.80	3.70	3.50	3.00	2.90							3.38
Argentina	AM	2.50	2.80	2.90	2.90	2.90	2.90	2.90	3.00	3.50	3.40	3.40	3.01
Dominican Republic	AM	2.90	3.00	2.80	3.00	3.00	3.00	3.00	2.60	3.20	2.90	3.20	2.96
Bolivia	AM	2.20	2.50	2.70	2.90	3.00	2.70	2.80	2.80	3.40	3.40	3.50	2.90

Guatemala	AM	2.20	2.50	2.60	2.80	3.10	3.40	3.20	2.70	3.30	2.90	3.20	2.90
Guyana	AM		2.50	2.50	2.60	2.60	2.60	2.70	2.50	2.80	2.70	3.00	2.65
Nicaragua	AM	2.70	2.60	2.60	2.60	2.50	2.50	2.50	2.50	2.90	2.80	2.80	2.64
Ecuador	AM	2.40	2.50	2.30	2.10	2.00	2.20	2.50	2.70	3.20	3.50	3.30	2.61
Honduras	AM	2.30	2.60	2.50	2.50	2.60	2.50	2.40	2.60	2.80	2.60	2.90	2.57
Paraguay	AM	1.90	2.10	2.60	2.40	2.40	2.10	2.20	2.20	2.50	2.40	2.40	2.29
Venezuela	AM	2.30	2.30	2.30	2.00	1.90	1.90	2.00	1.90	1.90	2.00	1.90	2.04
Haiti	AM	1.50	1.80	1.80	1.60	1.40	1.80	2.20	1.80	1.90	1.90	1.90	1.78
New Zealand	AP	9.60	9.60	9.60	9.40	9.30	9.40	9.30	9.50	9.00	9.10	9.10	9.35
Singapore	AP	9.30	9.40	9.40	9.30	9.20	9.20	9.30	9.20	8.70	8.60	8.40	9.09
Australia	AP	8.80	8.80	8.70	8.60	8.70	8.70	8.70	8.80	8.50	8.10	8.00	8.58
Hong Kong	AP	8.00	8.30	8.30	8.30	8.10	8.20	8.40	8.40	7.70	7.50	7.40	8.05
Japan	AP	6.90	7.30	7.60	7.50	7.30	7.70	7.80	8.00	7.40	7.40	7.60	7.50
Taiwan	AP	5.60	5.90	5.90	5.70	5.70	5.60	5.80	6.10	6.10	6.10	6.10	5.87
Bhutan	AP			6.00	5.00	5.20	5.00	5.70	5.70	6.30	6.30	6.50	5.74
Brunei Darussalam	AP						5.50	5.50	5.20	5.50	6.00		5.54
Macau	AP			6.60	5.70	5.40	5.30	5.00	5.10				5.52
Korea (South)	AP		4.50	5.00	5.10		5.50	5.40	5.40	5.60	5.50	5.50	5.28
Malaysia	AP	5.00	5.10	5.00	5.10	5.10	4.50	4.40	4.30	4.90	5.00	5.20	4.87
Samoa	AP				4.50	4.40	4.50	4.10	3.90				4.28
Fiji	AP		4.00										4.00
China	AP	3.40	3.20	3.30	3.50	3.60	3.60	3.50	3.60	3.90	4.00	3.60	3.56
Thailand	AP	3.60	3.80	3.60	3.30	3.50	3.40	3.50	3.40	3.70	3.50	3.80	3.55
Sri Lanka	AP	3.50	3.20	3.10	3.20	3.20	3.10	3.20	3.30	4.00	3.70	3.80	3.39
India	AP	2.80	2.90	3.30	3.50	3.40	3.40	3.30	3.10	3.60	3.60	3.80	3.34
Vanuatu	AP				3.10	2.90	3.20	3.60	3.50				3.26
Mongolia	AP	3.00	3.00	2.80	3.00	3.00	2.70	2.70	2.70	3.60	3.80	3.90	3.11
Kiribati	AP				3.30	3.10	2.80	3.20	3.10				3.10
Solomon Islands	AP				2.80	2.90	2.80	2.80	2.70				2.80
Vietnam	AP	2.60	2.60	2.60	2.60	2.70	2.70	2.70	2.90	3.10	3.10	3.10	2.79
Philippines	AP	2.60	2.50	2.50	2.50	2.30	2.40	2.40	2.60	3.40	3.60	3.80	2.78
Indonesia	AP	2.00	2.20	2.40	2.30	2.60	2.80	2.80	3.00	3.20	3.20	3.40	2.72
Maldives	AP				3.30	2.80	2.50	2.30	2.50				2.68
Tonga	AP				1.70	2.40	3.00	3.00	3.10				2.64
Timor-Leste	AP			2.60	2.60	2.20	2.20	2.50	2.40	3.30	3.00	2.80	2.62
Nepal	AP	2.80	2.50	2.50	2.50	2.70	2.30	2.20	2.20	2.70	3.10	2.90	2.58
Pakistan	AP	2.10	2.10	2.20	2.40	2.50	2.40	2.30	2.50	2.70	2.80	2.90	2.45
Laos	AP		3.30	2.60	1.90	2.00	2.00	2.10	2.20	2.10	2.60	2.50	2.33
Papua New Guinea	AP	2.60	2.30	2.40	2.00	2.00	2.10	2.10	2.20	2.50	2.50	2.50	2.29
Bangladesh	AP	1.50	1.70	2.00	2.00	2.10	2.40	2.40	2.70	2.60	2.70	2.50	2.24
Cambodia	AP		2.30	2.10	2.00	1.80	2.00	2.10	2.10	2.20	2.00	2.10	2.07
Myanmar	AP	1.70	1.80	1.90	1.40	1.30	1.40	1.40	1.50	1.50	2.10	2.10	1.65
Afghanistan	AP		2.50		1.80	1.50	1.30	1.40	1.50	0.80	0.80	1.20	1.42
Korea (North)	AP								1.00	0.80	0.80	0.80	0.85
Turkey	EE	3.20	3.50	3.80	4.10	4.60	4.40	4.40	4.20	4.90	5.00	4.50	4.24
Georgia	EE	2.00	2.30	2.80	3.40	3.90	4.10	3.80	4.10	5.20	4.90	5.20	3.79
Montenegro	EE	2.70	2.80		3.30	3.40	3.90	3.70	4.00	4.10	4.40	4.20	3.65
Serbia	EE	2.70	2.80	3.00	3.40	3.40	3.50	3.50	3.30	3.90	4.20	4.10	3.44
Bosnia and Herzegovina	EE	3.10	2.90	2.90	3.30	3.20	3.00	3.20	3.20	4.20	4.20	3.90	3.37
Macedonia	EE	2.70	2.70	2.70	3.30	3.60	3.80					4.50	3.33
Kosovo	EE							2.80	2.90	3.40	3.30	3.30	3.14
Moldova	EE	2.30	2.90	3.20	2.80	2.90	3.30	2.90	2.90	3.60	3.50	3.50	3.07
Armenia	EE	3.10	2.90	2.90	3.00	2.90	2.70	2.60	2.60	3.40	3.60	3.70	3.04
Albania	EE	2.50	2.40	2.60	2.90	3.40	3.20	3.30	3.10	3.30	3.10	3.30	3.01
Belarus	EE	3.30	2.60	2.10	2.10	2.00	2.40	2.50	2.40	3.10	2.90	3.10	2.59
Kazakhstan	EE	2.20	2.60	2.60	2.10	2.20	2.70	2.90	2.70	2.80	2.60	2.90	2.57
Ukraine	EE	2.20	2.60	2.80	2.70	2.50	2.20	2.40	2.30	2.60	2.50	2.60	2.49
Russia	EE	2.80	2.40	2.50	2.30	2.10	2.20	2.10	2.40	2.80	2.80	2.70	2.46
Azerbaijan	EE	1.90	2.20	2.40	2.10	1.90	2.30	2.40	2.40	2.70	2.80	2.90	2.36
Kyrgyzstan	EE	2.20	2.30	2.20	2.10	1.80	1.90	2.00	2.10	2.40	2.40	2.70	2.19
Tajikistan	EE	2.00	2.10	2.20	2.10	2.00	2.00	2.10	2.30	2.20	2.20	2.30	2.14
Uzbekistan	EE	2.30	2.20	2.10	1.70	1.80	1.70	1.60	1.60	1.70	1.70	1.80	1.84
Turkmenistan	EE	2.00	1.80	2.20	2.00	1.80	1.80	1.60	1.60	1.70	1.70	1.70	1.81
Denmark	EU	9.50	9.50	9.50	9.40	9.30	9.30	9.30	9.40	9.00	9.10	9.20	9.32
Finland	EU	9.70	9.60	9.60	9.40	9.00	8.90	9.20	9.40	9.00	8.90	8.90	9.24
Sweden	EU	9.20	9.20	9.20	9.30	9.30	9.20	9.20	9.30	8.80	8.90	8.70	9.12
Switzerland	EU	9.10	9.10	9.10	9.00	9.00	9.00	8.70	8.80	8.60	8.50	8.60	8.86
Iceland	EU	9.50	9.70	9.60	9.20	8.90	8.70	8.50	8.30	8.20	7.80	7.90	8.75
Netherlands	EU	8.70	8.60	8.70	9.00	8.90	8.90	8.80	8.90	8.40	8.30	8.30	8.68
Norway	EU	8.90	8.90	8.80	8.70	7.90	8.60	8.60	9.00	8.50	8.60	8.60	8.65
Luxembourg	EU	8.40	8.50	8.60	8.40	8.30	8.20	8.50	8.50	8.00	8.00	8.20	8.33
United Kingdom	EU	8.60	8.60	8.60	8.40	7.70	7.70	7.60	7.80	7.40	7.60	7.80	7.98
Germany	EU	8.20	8.20	8.00	7.80	7.90	8.00	7.90	8.00	7.90	7.80	7.90	7.96
Austria	EU	8.40	8.70	8.60	8.10	8.10	7.90	7.90	7.80	6.90	6.90	7.20	7.86
Ireland	EU	7.50	7.40	7.40	7.50	7.70	8.00	8.00	7.50	6.90	7.20	7.40	7.50
Belgium	EU	7.50	7.40	7.30	7.10	7.30	7.10	7.10	7.50	7.50	7.50	7.60	7.35
France	EU	7.10	7.50	7.40	7.30	6.90	6.90	6.80	7.00	7.10	7.10	6.90	7.09

Estonia	EU	6.00	6.40	6.70	6.50	6.60	6.60	6.50	6.40	6.40	6.80	6.90	6.53
Spain	EU	7.10	7.00	6.80	6.70	6.50	6.10	6.10	6.20	6.50	5.90	6.00	6.45
Portugal	EU	6.30	6.50	6.60	6.50	6.10	5.80	6.00	6.10	6.30	6.20	6.30	6.25
Slovenia	EU	6.00	6.10	6.40	6.60	6.70	6.60	6.40	5.90	6.10	5.70	5.80	6.21
Cyprus	EU	5.40	5.70	5.60	5.30	6.40	6.60	6.30	6.30	6.60	6.30	6.30	6.07
Malta	EU	6.80	6.60	6.40	5.80	5.80	5.20	5.60	5.60	5.70	5.60	5.50	5.87
Hungary	EU	4.80	5.00	5.20	5.30	5.10	5.10	4.70	4.60	5.50	5.40	5.40	5.10
Lithuania	EU	4.60	4.80	4.80	4.80	4.60	4.90	5.00	4.80	5.40	5.70	5.80	5.02
Poland	EU	3.50	3.40	3.70	4.20	4.60	5.00	5.30	5.50	5.80	6.00	6.10	4.83
Czech Republic	EU	4.20	4.30	4.80	5.20	5.20	4.90	4.60	4.40	4.90	4.80	5.10	4.76
Latvia	EU	4.00	4.20	4.70	4.80	5.00	4.50	4.30	4.20	4.90	5.30	5.50	4.67
Slovakia	EU	4.00	4.30	4.70	4.90	5.00	4.50	4.30	4.00	4.60	4.70	5.00	4.55
Italy	EU	4.80	5.00	4.90	5.20	4.80	4.30	3.90	3.90	4.20	4.30	4.30	4.51
Croatia	EU	3.50	3.40	3.40	4.10	4.40	4.10	4.10	4.00	4.60	4.80	4.80	4.11
Greece	EU	4.30	4.30	4.40	4.60	4.70	3.80	3.50	3.40	3.60	4.00	4.30	4.08
Bulgaria	EU	4.10	4.00	4.00	4.10	3.60	3.80	3.60	3.30	4.10	4.10	4.30	3.91
Romania	EU	2.90	3.00	3.10	3.70	3.80	3.80	3.70	3.60	4.40	4.30	4.30	3.69
Qatar	ME	5.20	5.90	6.00	6.00	6.50	7.00	7.70	7.20	6.80	6.80	6.90	6.55
United Arab Emirates	ME	6.10	6.20	6.20	5.70	5.90	6.50	6.30	6.80	6.80	6.90	7.00	6.40
Israel	ME	6.40	6.30	5.90	6.10	6.00	6.10	6.10	5.80	6.00	6.10	6.00	6.07
Bahrain	ME	5.80	5.80	5.70	5.00	5.40	5.10	4.90	5.10	5.10	4.80	4.90	5.24
Oman	ME	6.10	6.30	5.40	4.70	5.50	5.50	5.30	4.80	4.70	4.70	4.50	5.23
Jordan	ME	5.30	5.70	5.30	4.70	5.10	5.00	4.70	4.50	4.80	4.50	4.90	4.95
Kuwait	ME	4.60	4.70	4.80	4.30	4.30	4.10	4.50	4.60	4.40	4.30	4.40	4.45
Tunisia	ME	5.00	4.90	4.60	4.20	4.40	4.20	4.30	3.80	4.10	4.10	4.00	4.33
Saudi Arabia	ME	3.40	3.40	3.30	3.40	3.50	4.30	4.70	4.40	4.40	4.60	4.90	4.03
Morocco	ME	3.20	3.20	3.20	3.50	3.50	3.30	3.40	3.40	3.70	3.70	3.90	3.45
Egypt	ME	3.20	3.40	3.30	2.90	2.80	2.80	3.10	2.90	3.20	3.20	3.70	3.14
Algeria	ME	2.70	2.80	3.10	3.00	3.20	2.80	2.90	2.90	3.40	3.60	3.60	3.09
Lebanon	ME	2.70	3.10	3.60	3.00	3.00	2.50	2.50	2.50	3.00	2.80	2.70	2.85
Syria	ME	3.40	3.40	2.90	2.40	2.10	2.60	2.50	2.60	2.60	1.70	2.00	2.56
Palestine	ME	2.50	2.60										2.55
Iran	ME	2.90	2.90	2.70	2.50	2.30	1.80	2.20	2.70	2.80	2.50	2.70	2.55
Libya	ME	2.50	2.50	2.70	2.50	2.60	2.50	2.20	2.00	2.10	1.50	1.80	2.26
Yemen	ME	2.40	2.70	2.60	2.50	2.30	2.10	2.20	2.10	2.30	1.80	1.90	2.26
Iraq	ME	2.10	2.20	1.90	1.50	1.30	1.50	1.50	1.80	1.80	1.60	1.60	1.71
Sudan	ME	2.20	2.10	2.00	1.80	1.60	1.50	1.60	1.60	1.30	1.10	1.10	1.63

This table reports the scores of the Corruption Perceptions Index (*CPI*) over the 2004-2014 period published annually by Transparency International. For scores in 2012, 2013, and 2014, the original scores range between 0 and 100; we divide these scores by 10 to be consistent with the values in the previous years that range between 0 and 10. “Area” column reports the regions to which each country or territory belongs as classified by Transparency International. “AF” stands for Africa, “AM” stands for America, “AP” stands for Asian Pacific, “EE” stands for Eastern Europe, “EU” stands for European Union, and “ME” stands for the Middle East. The last column reports the average value of *CPI* over the period and the values used in this study.

Appendix I.B: The Control of Corruption Index (CPI), 2004 - 2014

Country/Territory	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	<i>CCI</i>
Denmark	2.43	2.29	2.47	2.45	2.39	2.45	2.36	2.40	2.38	2.40	2.25	2.39
New Zealand	2.38	2.20	2.33	2.33	2.31	2.39	2.34	2.30	2.32	2.34	2.25	2.32
Finland	2.44	2.33	2.46	2.40	2.34	2.25	2.16	2.20	2.24	2.20	2.17	2.29
Sweden	2.16	2.02	2.20	2.24	2.23	2.25	2.27	2.20	2.31	2.29	2.15	2.21
Singapore	2.33	2.17	2.19	2.24	2.25	2.22	2.18	2.11	2.12	2.08	2.07	2.18
Switzerland	2.01	2.02	2.09	2.15	2.12	2.07	2.07	2.04	2.15	2.13	2.15	2.09
Norway	1.94	2.01	2.13	1.99	1.91	1.99	2.09	2.15	2.26	2.29	2.23	2.09
Netherlands	2.00	1.97	2.05	2.17	2.12	2.13	2.14	2.12	2.12	2.05	1.99	2.08
Iceland	2.19	2.31	2.18	2.21	2.34	2.04	1.94	1.94	1.89	1.91	1.83	2.07
Luxembourg	1.84	1.66	1.88	1.97	1.97	1.97	2.05	2.16	2.12	2.12	2.07	1.98
Australia	2.03	1.95	1.96	2.01	2.04	2.05	2.03	2.04	1.99	1.79	1.85	1.98
Canada	1.83	1.88	1.96	2.00	2.00	2.06	2.07	1.98	1.93	1.89	1.84	1.95
Hong Kong SAR, China	1.87	1.78	1.87	1.92	1.90	1.89	1.96	1.85	1.74	1.64	1.62	1.82
Germany	1.86	1.89	1.80	1.74	1.76	1.76	1.78	1.74	1.83	1.81	1.84	1.80
United Kingdom	1.93	1.90	1.79	1.74	1.68	1.63	1.60	1.62	1.67	1.70	1.74	1.73
Austria	2.05	1.92	1.91	2.01	1.84	1.70	1.59	1.43	1.39	1.55	1.47	1.72
Ireland	1.29	1.59	1.71	1.75	1.75	1.76	1.69	1.56	1.46	1.54	1.60	1.61
Liechtenstein	1.37	1.28	1.26	1.09	1.19	1.69	1.85	1.81	1.80	1.84	2.08	1.57
Belgium	1.38	1.39	1.31	1.36	1.37	1.46	1.53	1.58	1.61	1.67	1.57	1.47
Chile	1.37	1.47	1.46	1.40	1.38	1.38	1.50	1.53	1.58	1.54	1.49	1.46
Barbados	1.43	1.42	1.33	1.41	1.37	1.38	1.48	1.72	1.66	1.62	1.14	1.45
Japan	1.22	1.22	1.33	1.24	1.34	1.38	1.56	1.56	1.63	1.66	1.69	1.44
France	1.33	1.37	1.46	1.46	1.41	1.44	1.47	1.53	1.46	1.33	1.31	1.42
United States	1.83	1.55	1.35	1.39	1.45	1.29	1.27	1.27	1.41	1.31	1.38	1.41
Bahamas, The	1.43	1.34	1.35	1.34	1.34	1.37	1.35	1.35	1.32	1.34	1.31	1.35
Bermuda	1.33	1.28	1.26	1.27	1.29	1.33	1.31	1.28	1.27	1.26	1.23	1.28
Andorra	1.14	1.28	1.26	1.27	1.29	1.33	1.31	1.28	1.27	1.26	1.23	1.26
Anguilla	0.86	1.28	1.26	1.27	1.29	1.33	1.31	1.28	1.27	1.26	1.23	1.24
Uruguay	0.90	1.07	1.11	1.24	1.31	1.24	1.28	1.28	1.36	1.38	1.38	1.23
Jersey, Channel Islands	..	..	..	..	..	..	..	1.23	1.22	1.24	1.21	1.23
Greenland	..	..	..	..	..	1.19	1.22	1.23	1.22	1.24	1.21	1.22
Cayman Islands	1.28	1.28	1.26	1.27	0.81	1.12	1.13	1.35	1.35	1.36	1.02	1.20
Aruba	1.18	1.28	1.26	1.27	1.29	1.12	1.13	1.11	1.10	1.13	1.02	1.17
Antigua and Barbuda	0.86	0.75	1.13	1.15	1.29	1.33	1.31	1.28	1.27	1.26	0.64	1.11
Spain	1.36	1.34	1.19	1.09	1.19	1.06	1.08	1.10	1.13	0.90	0.63	1.10
Estonia	0.98	1.03	1.02	1.00	0.99	1.01	1.00	1.05	1.10	1.19	1.30	1.06
United Arab Emirates	1.06	1.00	0.89	1.01	1.08	0.91	0.90	1.08	1.16	1.28	1.20	1.05
Cyprus	0.89	0.91	1.09	1.08	1.20	0.91	0.97	0.87	1.25	1.25	1.08	1.05
Portugal	1.14	1.07	1.02	1.04	1.07	1.09	1.09	1.11	0.96	0.95	0.95	1.04
Qatar	0.52	0.71	0.93	0.68	0.94	1.57	1.41	1.01	1.06	1.11	0.99	0.99
St. Lucia	0.31	1.12	1.05	1.06	1.16	1.22	1.20	1.21	1.01	1.04	0.41	0.98
Botswana	0.90	1.16	0.97	1.00	1.04	0.96	1.03	1.00	0.93	0.92	0.85	0.98
Netherlands Antilles	0.62	1.28	1.26	1.27	1.29	0.84	0.82	0.80	0.79	0.79	..	0.97
French Guiana	0.81	0.80	0.78	0.79	0.81	1.12	1.13	1.11	1.10	1.13	1.02	0.96
Slovenia	1.03	0.91	1.03	1.01	0.98	1.06	0.92	0.95	0.84	0.73	0.73	0.93
Bhutan	0.80	0.82	0.82	0.88	0.86	0.91	0.91	0.84	0.94	0.90	1.28	0.90
Malta	0.85	0.82	1.01	1.02	1.00	0.77	0.79	0.77	0.94	0.98	0.85	0.89
St. Vincent and the Grenadines	0.31	0.98	0.88	0.89	0.98	1.04	1.01	1.02	0.94	0.94	0.62	0.87
Israel	0.93	0.83	1.01	0.85	0.90	0.81	0.76	0.80	0.91	0.91	0.87	0.87
Martinique	0.90	0.80	0.78	0.79	0.81	0.84	0.82	0.80	0.79	0.79	1.23	0.85
St. Kitts and Nevis	0.26	0.98	0.88	0.89	0.98	1.04	1.01	1.02	0.94	0.94	0.25	0.83
Cabo Verde	0.46	0.56	0.81	0.89	0.87	0.87	0.86	0.90	0.87	0.84	0.94	0.81
Reunion	1.00	0.80	0.78	0.79	0.81	0.84	0.82	0.80	0.79	0.79	0.64	0.80
Guam	0.38	0.80	0.78	0.79	0.81	0.84	0.82	0.80	0.79	0.79	1.23	0.80
Virgin Islands (U.S.)	0.67	0.80	0.78	0.79	0.81	0.84	0.82	0.80	0.79	0.79	0.64	0.77
Taiwan, China	0.83	0.75	0.53	0.53	0.52	0.62	0.74	0.87	0.72	0.68	0.81	0.69
Dominica	0.61	0.66	0.62	0.62	0.68	0.73	0.71	0.71	0.66	0.67	0.62	0.66
Puerto Rico	1.28	1.28	0.51	0.41	0.42	0.49	0.48	0.44	0.54	0.47	0.46	0.62
Costa Rica	0.42	0.48	0.44	0.52	0.55	0.75	0.71	0.65	0.62	0.63	0.76	0.59
Macao SAR, China	1.42	0.52	0.35	0.45	0.44	0.16	0.42	0.44	0.43	0.55	0.85	0.55
Brunei Darussalam	0.41	0.25	0.21	0.21	0.49	0.99	0.86	0.86	0.55	0.65	0.53	0.55
Korea, Rep.	0.39	0.62	0.35	0.60	0.47	0.54	0.47	0.53	0.54	0.61	0.55	0.51
American Samoa	0.81	0.80	0.31	0.31	0.33	0.35	0.33	0.32	0.31	0.31	1.23	0.49
Hungary	0.69	0.65	0.66	0.62	0.47	0.43	0.37	0.40	0.36	0.32	0.16	0.47
Grenada	0.61	0.66	0.63	0.38	0.32	0.36	0.43	0.42	0.42	0.42	0.25	0.44
Poland	0.14	0.27	0.27	0.30	0.46	0.45	0.50	0.56	0.66	0.60	0.64	0.44
Mauritius	0.25	0.32	0.35	0.44	0.52	0.54	0.56	0.54	0.30	0.29	0.35	0.40
Czech Republic	0.44	0.49	0.36	0.30	0.36	0.39	0.33	0.34	0.27	0.23	0.37	0.35
Seychelles	0.27	0.23	0.12	0.19	0.39	0.43	0.45	0.42	0.42	0.44	0.44	0.35
Oman	0.67	0.39	0.35	0.35	0.52	0.33	0.32	0.14	0.18	0.16	0.31	0.34
Lithuania	0.43	0.33	0.16	0.13	0.14	0.23	0.38	0.33	0.39	0.43	0.56	0.32
Namibia	0.16	0.20	0.24	0.32	0.61	0.30	0.34	0.31	0.34	0.33	0.29	0.31
Latvia	0.18	0.38	0.37	0.35	0.25	0.23	0.23	0.29	0.25	0.33	0.42	0.30
Slovak Republic	0.39	0.49	0.42	0.35	0.36	0.27	0.29	0.28	0.10	0.08	0.16	0.29

Bahrain	0.45	0.39	0.19	0.18	0.19	0.19	0.18	0.22	0.37	0.43	0.28	0.28
Cuba	0.23	0.28	0.26	0.31	0.27	0.28	0.31	0.27	0.24	0.12	0.07	0.24
Kuwait	0.79	0.51	0.43	0.37	0.42	0.31	0.30	0.09	-0.19	-0.19	-0.24	0.24
Vanuatu	-0.32	0.24	0.17	0.23	0.29	0.30	0.29	0.30	0.34	0.29	0.46	0.23
Italy	0.38	0.41	0.48	0.34	0.27	0.20	0.13	0.18	0.07	0.05	-0.03	0.22
South Africa	0.46	0.57	0.45	0.25	0.21	0.18	0.13	0.06	-0.12	-0.07	-0.06	0.19
Malaysia	0.34	0.20	0.23	0.19	-0.05	-0.06	0.09	0.03	0.24	0.35	0.41	0.18
Jordan	0.26	0.26	0.26	0.26	0.36	0.16	0.04	0.10	0.07	0.07	0.14	0.18
Samoa	0.13	0.18	0.24	0.25	0.21	0.12	0.12	0.10	0.11	0.15	0.30	0.17
Rwanda	-0.44	-0.62	-0.21	0.00	0.10	0.09	0.35	0.36	0.56	0.63	0.76	0.14
Greece	0.46	0.36	0.36	0.27	0.13	0.07	-0.06	-0.10	-0.19	-0.05	-0.12	0.10
Croatia	0.26	0.17	0.08	0.09	-0.01	-0.05	0.06	0.06	0.01	0.12	0.22	0.09
Cook Islands	0.11	0.57	0.60	1.05	..	-0.58	-0.25	-0.23	-0.22	-0.22	..	0.09
Lesotho	-0.10	0.02	0.00	-0.07	0.05	0.16	0.17	0.15	0.12	0.28	0.19	0.09
Georgia	-0.47	-0.22	0.04	-0.13	-0.11	-0.12	0.01	0.12	0.40	0.47	0.79	0.07
Kiribati	0.32	0.20	0.05	0.10	-0.03	-0.16	-0.08	0.09	0.01	-0.04	0.25	0.07
Turkey	-0.18	-0.03	0.03	0.11	0.11	0.09	0.03	0.04	0.16	0.09	-0.15	0.03
Brazil	0.02	-0.14	-0.11	-0.08	0.01	-0.07	0.05	0.17	-0.04	-0.08	-0.34	-0.06
Saudi Arabia	-0.29	-0.10	-0.19	-0.17	-0.01	-0.02	0.04	-0.31	-0.04	-0.02	0.09	-0.09
Ghana	-0.25	-0.37	-0.02	0.04	-0.07	-0.01	0.01	-0.03	-0.13	-0.10	-0.19	-0.10
Fiji	0.16	-0.30	0.01	-0.05	-0.05	-0.34	-0.46	-0.13	-0.11	-0.12	0.18	-0.11
Trinidad and Tobago	-0.03	0.02	-0.11	-0.04	-0.06	-0.04	-0.17	-0.11	-0.16	-0.22	-0.41	-0.12
Micronesia, Fed. Sts.	-0.04	-0.24	-0.25	-0.34	-0.31	-0.13	-0.15	-0.34	-0.15	-0.20	0.77	-0.13
Suriname	0.28	0.22	-0.10	-0.01	-0.02	-0.26	-0.33	-0.28	-0.33	-0.28	-0.44	-0.14
Tunisia	0.08	-0.26	-0.19	-0.25	-0.30	-0.22	-0.26	-0.06	-0.06	-0.07	-0.04	-0.15
Bulgaria	0.11	0.06	-0.07	-0.18	-0.25	-0.21	-0.19	-0.22	-0.23	-0.27	-0.25	-0.15
Tuvalu	0.64	-0.11	-0.10	-0.24	-0.21	-0.22	-0.26	-0.53	-0.35	-0.40	-0.01	-0.16
Macedonia, FYR	-0.50	-0.45	-0.37	-0.36	-0.19	-0.13	-0.08	-0.09	-0.04	-0.05	-0.02	-0.21
Romania	-0.30	-0.24	-0.16	-0.18	-0.14	-0.26	-0.23	-0.21	-0.26	-0.19	-0.11	-0.21
Belize	-0.29	-0.30	-0.36	-0.37	-0.34	-0.05	-0.11	-0.33	-0.14	-0.10	-0.20	-0.23
Sao Tome and Principe	-0.36	-0.54	-0.29	-0.30	-0.30	-0.20	-0.25	-0.18	-0.18	-0.18	0.00	-0.25
West Bank and Gaza	-0.01	-0.49	-0.30	-0.11	-0.44	-0.10	-0.11	-0.45	-0.25	-0.25	-0.28	-0.25
Montenegro	-0.46	-0.35	-0.40	-0.33	-0.21	-0.20	-0.23	-0.20	-0.11	-0.25	-0.06	-0.25
Colombia	-0.13	-0.13	-0.12	-0.22	-0.24	-0.31	-0.39	-0.29	-0.39	-0.41	-0.37	-0.27
Jamaica	-0.35	-0.31	-0.26	-0.30	-0.30	-0.28	-0.23	-0.16	-0.26	-0.28	-0.30	-0.27
Nauru	..	..	..	-0.35	-0.32	-0.40	0.01	-0.04	-0.02	-0.64	-0.47	-0.28
Sri Lanka	-0.15	-0.34	-0.20	-0.15	-0.23	-0.40	-0.42	-0.39	-0.24	-0.21	-0.34	-0.28
Panama	-0.27	-0.35	-0.32	-0.30	-0.09	-0.30	-0.32	-0.32	-0.36	-0.35	-0.34	-0.30
Solomon Islands	-0.37	0.04	-0.22	-0.39	-0.37	-0.29	-0.35	-0.34	-0.36	-0.40	-0.28	-0.30
Peru	-0.32	-0.33	-0.20	-0.25	-0.19	-0.31	-0.23	-0.22	-0.37	-0.42	-0.56	-0.31
Senegal	-0.05	-0.05	-0.37	-0.51	-0.50	-0.49	-0.64	-0.50	-0.26	-0.19	0.06	-0.32
Bosnia and Herzegovina	-0.34	-0.24	-0.30	-0.38	-0.36	-0.38	-0.34	-0.32	-0.30	-0.24	-0.31	-0.32
Morocco	-0.14	-0.31	-0.41	-0.34	-0.38	-0.33	-0.20	-0.40	-0.44	-0.37	-0.27	-0.33
Serbia	-0.50	-0.41	-0.29	-0.35	-0.31	-0.31	-0.29	-0.27	-0.33	-0.30	-0.23	-0.33
Thailand	-0.23	-0.19	-0.38	-0.37	-0.41	-0.31	-0.33	-0.32	-0.37	-0.34	-0.45	-0.34
Burkina Faso	-0.14	-0.13	-0.31	-0.33	-0.30	-0.34	-0.35	-0.37	-0.48	-0.52	-0.46	-0.34
Eswatini	-0.53	-0.47	-0.22	-0.24	-0.22	-0.23	-0.23	-0.35	-0.41	-0.41	-0.43	-0.34
El Salvador	-0.42	-0.45	-0.30	-0.37	-0.36	-0.25	-0.28	-0.26	-0.41	-0.35	-0.38	-0.35
Mexico	-0.30	-0.27	-0.26	-0.27	-0.24	-0.30	-0.36	-0.40	-0.41	-0.51	-0.76	-0.37
Marshall Islands	-0.57	-0.46	-0.55	-0.61	-0.57	-0.34	-0.33	-0.33	-0.26	-0.04	-0.11	-0.38
Madagascar	-0.28	-0.07	-0.19	-0.19	-0.27	-0.34	-0.43	-0.48	-0.67	-0.75	-0.84	-0.41
Argentina	-0.42	-0.39	-0.34	-0.34	-0.44	-0.44	-0.36	-0.37	-0.44	-0.43	-0.54	-0.41
India	-0.41	-0.36	-0.28	-0.40	-0.34	-0.45	-0.47	-0.54	-0.51	-0.52	-0.43	-0.43
Zambia	-0.59	-0.59	-0.55	-0.41	-0.39	-0.46	-0.51	-0.40	-0.28	-0.31	-0.34	-0.44
Niue	..	..	..	..	..	-0.58	-0.46	-0.44	-0.44	-0.42	..	-0.47
Palau	..	..	..	..	-0.32	-0.40	-0.50	-0.53	-0.43	-0.64	-0.47	-0.47
China	-0.56	-0.61	-0.51	-0.59	-0.52	-0.51	-0.56	-0.51	-0.44	-0.36	-0.34	-0.50
Djibouti	-0.58	-0.71	-0.66	-0.54	-0.33	-0.37	-0.40	-0.41	-0.46	-0.54	-0.60	-0.51
Mozambique	-0.60	-0.53	-0.62	-0.52	-0.49	-0.44	-0.45	-0.49	-0.57	-0.60	-0.67	-0.54
Tanzania	-0.56	-0.60	-0.23	-0.34	-0.42	-0.45	-0.54	-0.59	-0.76	-0.77	-0.75	-0.55
Eritrea	-0.22	-0.36	-0.35	-0.44	-0.39	-0.49	-0.56	-0.63	-0.78	-0.90	-0.89	-0.55
Algeria	-0.68	-0.48	-0.52	-0.56	-0.59	-0.58	-0.52	-0.54	-0.50	-0.47	-0.60	-0.55
Malawi	-0.77	-0.71	-0.55	-0.51	-0.45	-0.40	-0.49	-0.42	-0.47	-0.61	-0.75	-0.56
Kosovo	-0.29	-0.52	-0.51	-0.73	-0.63	-0.58	-0.62	-0.61	-0.65	-0.65	-0.49	-0.57
Bolivia	-0.78	-0.74	-0.43	-0.40	-0.52	-0.63	-0.47	-0.57	-0.73	-0.59	-0.63	-0.59
Mongolia	-0.40	-0.61	-0.58	-0.61	-0.66	-0.76	-0.74	-0.69	-0.54	-0.48	-0.47	-0.59
Guyana	-0.45	-0.56	-0.58	-0.59	-0.56	-0.54	-0.56	-0.61	-0.75	-0.65	-0.74	-0.60
Maldives	-0.35	-0.44	-0.60	-0.86	-0.82	-0.67	-0.55	-0.55	-0.70	-0.75	-0.38	-0.61
Belarus	-0.80	-0.75	-0.62	-0.66	-0.63	-0.63	-0.69	-0.68	-0.52	-0.47	-0.30	-0.61
Vietnam	-0.73	-0.72	-0.75	-0.63	-0.71	-0.54	-0.62	-0.61	-0.53	-0.48	-0.44	-0.61
Mali	-0.59	-0.50	-0.48	-0.43	-0.53	-0.67	-0.68	-0.64	-0.83	-0.79	-0.75	-0.62
Benin	-0.46	-0.85	-0.59	-0.46	-0.50	-0.62	-0.66	-0.59	-0.86	-0.74	-0.67	-0.64
Armenia	-0.61	-0.67	-0.67	-0.74	-0.71	-0.62	-0.70	-0.66	-0.59	-0.53	-0.52	-0.64
Ethiopia	-0.73	-0.77	-0.65	-0.63	-0.66	-0.70	-0.69	-0.66	-0.60	-0.49	-0.44	-0.64
Tonga	-0.43	-1.33	-1.34	-1.04	-0.66	-0.68	-0.33	-0.33	-0.32	-0.28	-0.31	-0.64
Timor-Leste	-0.30	-0.56	-0.60	-0.65	-0.65	-0.80	-0.76	-0.81	-0.74	-0.68	-0.53	-0.64
Guatemala	-0.61	-0.67	-0.81	-0.74	-0.65	-0.53	-0.53	-0.53	-0.66	-0.62	-0.74	-0.64

Philippines	-0.65	-0.62	-0.83	-0.72	-0.76	-0.77	-0.76	-0.67	-0.56	-0.38	-0.44	-0.65
Gambia, The	-0.61	-0.71	-0.72	-0.73	-0.74	-0.57	-0.59	-0.51	-0.64	-0.70	-0.69	-0.66
Egypt, Arab Rep.	-0.65	-0.62	-0.75	-0.76	-0.78	-0.52	-0.63	-0.70	-0.60	-0.63	-0.62	-0.66
Albania	-0.70	-0.79	-0.80	-0.69	-0.59	-0.54	-0.53	-0.68	-0.73	-0.70	-0.55	-0.66
Iran, Islamic Rep.	-0.39	-0.48	-0.48	-0.55	-0.78	-0.84	-0.95	-0.87	-0.79	-0.69	-0.62	-0.68
Mauritania	-0.53	-0.56	-0.74	-0.58	-0.79	-0.61	-0.72	-0.63	-0.78	-0.83	-0.93	-0.70
Niger	-0.82	-0.72	-0.85	-0.75	-0.72	-0.61	-0.66	-0.64	-0.64	-0.61	-0.69	-0.70
Moldova	-0.98	-0.67	-0.64	-0.66	-0.63	-0.70	-0.67	-0.62	-0.61	-0.75	-0.85	-0.71
Nicaragua	-0.40	-0.61	-0.67	-0.77	-0.76	-0.72	-0.75	-0.73	-0.77	-0.74	-0.89	-0.71
Ecuador	-0.70	-0.71	-0.75	-0.77	-0.69	-0.80	-0.77	-0.71	-0.58	-0.57	-0.75	-0.71
Indonesia	-0.93	-0.88	-0.84	-0.62	-0.59	-0.84	-0.75	-0.70	-0.64	-0.61	-0.56	-0.72
Nepal	-0.87	-0.74	-0.69	-0.79	-0.81	-0.70	-0.69	-0.78	-0.81	-0.69	-0.59	-0.74
Liberia	-1.32	-1.20	-0.72	-0.46	-0.74	-0.59	-0.56	-0.67	-0.60	-0.71	-0.77	-0.76
Dominican Republic	-0.59	-0.69	-0.70	-0.75	-0.73	-0.78	-0.84	-0.81	-0.86	-0.87	-0.82	-0.77
Comoros	-0.90	-0.88	-0.70	-0.71	-0.79	-0.83	-0.81	-0.79	-0.78	-0.74	-0.61	-0.78
Lebanon	-0.66	-0.53	-0.94	-0.89	-0.82	-0.83	-0.88	-0.90	-0.87	-0.92	-1.04	-0.84
Gabon	-0.85	-0.67	-0.97	-1.06	-1.08	-1.00	-0.87	-0.89	-0.70	-0.66	-0.69	-0.86
Honduras	-0.89	-0.81	-0.83	-0.79	-0.90	-0.90	-0.89	-0.83	-0.95	-0.95	-0.79	-0.87
Uganda	-0.78	-0.82	-0.79	-0.82	-0.83	-0.90	-0.92	-0.92	-0.99	-1.04	-1.09	-0.90
Sierra Leone	-0.91	-1.07	-1.02	-0.91	-0.97	-0.93	-0.77	-0.84	-0.95	-0.91	-0.94	-0.93
Ukraine	-0.93	-0.72	-0.75	-0.80	-0.84	-1.04	-1.03	-1.05	-1.08	-1.13	-0.99	-0.94
Kazakhstan	-1.10	-1.01	-0.99	-0.99	-0.98	-0.93	-1.00	-0.99	-0.92	-0.93	-0.83	-0.97
Togo	-0.96	-0.87	-1.04	-0.99	-0.97	-1.02	-0.96	-1.01	-1.00	-1.03	-0.89	-0.98
Pakistan	-1.08	-1.05	-0.81	-0.82	-0.90	-1.07	-1.09	-1.08	-1.06	-0.96	-0.83	-0.98
Kenya	-0.86	-1.00	-0.93	-0.97	-1.06	-1.06	-0.91	-0.95	-1.09	-1.03	-0.93	-0.98
Russian Federation	-0.80	-0.82	-0.91	-1.01	-1.11	-1.13	-1.09	-1.07	-1.04	-1.01	-0.92	-0.99
Cote d'Ivoire	-1.21	-1.24	-1.20	-1.09	-1.11	-1.10	-1.16	-1.01	-0.83	-0.74	-0.42	-1.01
Paraguay	-1.37	-1.32	-1.14	-1.18	-0.98	-0.80	-0.73	-0.72	-0.86	-1.04	-0.97	-1.01
Papua New Guinea	-0.96	-1.13	-1.14	-1.11	-1.15	-1.23	-1.04	-1.02	-0.94	-0.96	-0.91	-1.05
Yemen, Rep.	-1.08	-0.89	-0.79	-0.79	-0.79	-1.07	-1.19	-1.23	-1.25	-1.25	-1.56	-1.08
Guinea	-0.92	-1.05	-1.12	-1.28	-1.20	-1.08	-1.20	-1.10	-1.04	-1.04	-1.06	-1.10
Bangladesh	-1.50	-1.39	-1.43	-1.06	-1.03	-1.07	-1.06	-1.09	-0.85	-0.89	-0.89	-1.11
Cameroon	-1.11	-1.18	-1.09	-1.02	-1.02	-1.00	-1.06	-1.15	-1.26	-1.21	-1.16	-1.11
Central African Republic	-1.39	-1.28	-1.21	-1.19	-1.12	-0.98	-0.93	-0.95	-0.99	-1.09	-1.16	-1.12
Azerbaijan	-1.16	-1.04	-1.09	-1.12	-1.15	-1.19	-1.24	-1.18	-1.13	-0.97	-1.02	-1.12
Syrian Arab Republic	-0.82	-0.85	-1.06	-1.10	-1.15	-1.13	-1.13	-1.09	-1.21	-1.26	-1.55	-1.12
Congo, Rep.	-0.93	-1.04	-1.10	-1.15	-1.18	-1.20	-1.15	-1.12	-1.21	-1.15	-1.18	-1.13
Nigeria	-1.34	-1.16	-1.12	-1.05	-0.89	-1.03	-1.05	-1.17	-1.17	-1.22	-1.27	-1.14
Lao PDR	-1.24	-1.30	-1.31	-1.27	-1.20	-1.24	-1.19	-1.18	-1.02	-0.93	-0.85	-1.16
Burundi	-0.89	-0.93	-1.10	-1.16	-1.09	-1.13	-1.17	-1.19	-1.45	-1.41	-1.26	-1.16
Cambodia	-1.06	-1.21	-1.25	-1.15	-1.23	-1.17	-1.24	-1.24	-1.07	-1.05	-1.14	-1.17
Venezuela, RB	-0.98	-1.03	-1.04	-1.11	-1.17	-1.19	-1.23	-1.17	-1.27	-1.31	-1.40	-1.17
Tajikistan	-1.29	-1.17	-1.03	-1.04	-1.18	-1.21	-1.29	-1.22	-1.28	-1.28	-1.13	-1.19
Uzbekistan	-1.13	-1.23	-1.00	-1.03	-1.06	-1.27	-1.30	-1.37	-1.30	-1.26	-1.19	-1.20
Libya	-0.91	-0.96	-1.08	-1.05	-0.95	-1.21	-1.29	-1.30	-1.36	-1.48	-1.56	-1.20
Kyrgyz Republic	-1.10	-1.25	-1.32	-1.32	-1.22	-1.31	-1.17	-1.22	-1.15	-1.16	-1.13	-1.21
Guinea-Bissau	-1.22	-1.15	-1.09	-1.19	-1.18	-1.20	-1.16	-1.16	-1.28	-1.33	-1.54	-1.23
Haiti	-1.47	-1.37	-1.31	-1.28	-1.21	-1.09	-1.19	-1.22	-1.24	-1.13	-1.23	-1.25
South Sudan	..	..	..	..	..	..	-0.82	-1.40	-1.28	-1.38	-1.59	-1.29
Sudan	-1.28	-1.37	-1.18	-1.32	-1.42	-1.15	-1.19	-1.18	-1.49	-1.47	-1.45	-1.32
Angola	-1.31	-1.30	-1.22	-1.29	-1.28	-1.40	-1.33	-1.34	-1.27	-1.31	-1.44	-1.32
Iraq	-1.48	-1.37	-1.45	-1.46	-1.46	-1.33	-1.26	-1.17	-1.22	-1.28	-1.33	-1.35
Korea, Dem. People's Rep.	-1.38	-1.28	-1.44	-1.54	-1.52	-1.29	-1.28	-1.31	-1.28	-1.28	-1.30	-1.36
Zimbabwe	-1.33	-1.31	-1.36	-1.40	-1.36	-1.36	-1.37	-1.40	-1.37	-1.40	-1.39	-1.37
Congo, Dem. Rep.	-1.45	-1.42	-1.53	-1.36	-1.24	-1.39	-1.44	-1.43	-1.31	-1.32	-1.30	-1.38
Chad	-1.38	-1.51	-1.35	-1.38	-1.52	-1.40	-1.38	-1.33	-1.31	-1.35	-1.32	-1.39
Myanmar	-1.65	-1.54	-1.66	-1.63	-1.62	-1.66	-1.67	-1.59	-1.06	-1.00	-0.88	-1.45
Turkmenistan	-1.47	-1.46	-1.52	-1.52	-1.50	-1.53	-1.53	-1.54	-1.43	-1.41	-1.31	-1.48
Afghanistan	-1.35	-1.44	-1.43	-1.59	-1.64	-1.53	-1.64	-1.58	-1.42	-1.44	-1.35	-1.49
Equatorial Guinea	-1.66	-1.61	-1.59	-1.50	-1.47	-1.47	-1.47	-1.46	-1.52	-1.59	-1.77	-1.56
Somalia	-1.79	-1.68	-1.73	-1.76	-1.87	-1.72	-1.74	-1.71	-1.59	-1.58	-1.66	-1.71

This table reports the scores of Control of Corruption Index (CCI) over the 2004-2014 period published annually by The World Bank. The last column report the average value of CCI over the period and the values used in this study.

Appendix II: Variable description

Variable	Description	Data source
Measures of stock returns		
Mean returns	The geometric mean of daily returns after subtracting the risk-free rate	Datastream
Abnormal returns	The geometric compound the daily abnormal returns after matching a firm's stock return with Fama and French's (2015) five factors as well as momentum (the average return on the two high prior return portfolios minus the average return on the two low prior return portfolios).	Datastream
Measures of corruption exposure		
CEI_CPI	The corruption exposure index constructed using the corruption perceptions index of Transparency International by a similar approach to Zeume (2017) as follows, $CEI_CPI_{i,t} = \sum_{c \in C} [(10 - CPI_c) \times \frac{NS_{i,c,t}}{TNS_{i,t}}]$ where $CEI_CPI_{i,t}$ is the corruption exposure index computed according to the CPI of firm i in year t, CPI_c is the average Transparency International's corruption perceptions index of country c over the sample period, $NS_{i,c,t}$ is the number of subsidiaries in country c owned by firm i in year t, $TNS_{i,t}$ is the total number of subsidiaries of firm i in year t.	Transparency International
CEI_CCI	The corruption exposure index constructed using the control of corruption index of the World Bank by a similar approach to Zeume (2017) as follows, $CEI_CCI_{i,t} = \sum_{c \in C} [(2.5 - CCI_c) \times \frac{NS_{i,c,t}}{TNS_{i,t}}]$ where and $CEI_CCI_{i,t}$ is the corruption exposure index computed according to the CCI of firm i in year t, CCI_c is the average the World Bank's Control of Corruption Index of country c over the sample period, $NS_{i,c,t}$ is the number of subsidiaries in country c owned by firm i in year t, $TNS_{i,t}$ is the total number of subsidiaries of firm i in year t.	World Bank
Measures of Africa exposure		
Has African subsidiary	A dummy variable equal to 1 if a firm has at least one subsidiary headquartered in Africa.	Orbis
ANTN	The ratio of the number of subsidiaries in Africa to the total number of subsidiaries of a firm.	Orbis
Country-level characteristics		
Egalitarianism distance	The distance in the degree of egalitarianism between two countries is calculated using the same approach of Siegel et al. (2011), where egalitarianism scores are from Schwartz (2014).	Schwartz (2014)
Difference in law (%)	The percentage of a firm's subsidiaries operating in the countries that have different law system from which of the United States (Common law).	Djankov et al. (2007)
Common language (%)	The percentage of a firm's subsidiaries operating in the countries that have the same language with which of the United States (English).	CEPII (Centre d'Etudes Prospectives et d'Informations Internationales)
Common religion (%)	The percentage of a firm's subsidiaries operating in the countries that have the same religion with which of the United States (Protestant).	Djankov et al. (2007)
Geographical distance (in million km)	The weighted average of geographical distances between the largest city of the United States (New York) and the largest city of each country where a firm has subsidiaries. The weight is the percentage of the number of subsidiaries in the given country to the total number of subsidiaries of the firm.	CEPII (Centre d'Etudes Prospectives et d'Informations Internationales)
Log of host country GDP	The average of the natural logarithm of country-level GDP.	World Bank
Corporate tax rate	The weighted average of corporate tax rates of countries where a firm has subsidiaries. The weight is the percentage of the number of subsidiaries in the given country to the total number of subsidiaries of the firm.	World Bank
Minor investors protection	The weighted average of the protecting minority investors scores, which measure the legal protection of minority shareholders against expropriation by corporate insiders. The weight is the percentage of the number of subsidiaries in the given country to the total number of subsidiaries of the firm.	Doing Business project of the World Bank
Underemployment benefits	The weighted average of scores of unemployment benefits, which account for the following four normalized variables: (1) the number of months of contributions or employment required to qualify for unemployment benefits by law; (2) the percentage of the worker's monthly salary deducted by law to cover unemployment benefits; (3) the waiting period for unemployment benefits; and (4) the percentage of the net salary covered by the net unemployment benefits in case of a one-year unemployment spell. The weight is the percentage of the number of subsidiaries in the given country to the total number of subsidiaries of the firm.	Botero et al. (2004)
Firm-level characteristics		
LnTA	The natural logarithm of total assets at the beginning of the calendar year.	Datastream
LnBTM	The natural logarithm of the book to market equity ratio at the beginning of the calendar year.	Datastream
Trading volume	The average daily number of shares traded in million.	Datastream
Price volatility	A measure of a stock's average annual price movement to a high and low from a mean price for each year.	Datastream
Ln(Intangible/Assets)	Book value of a firm's intangible assets scaled by the book value of its total assets.	Datastream

Ln(RD/Expense)	The natural logarithm of a fraction of a firm's R&D expenditures to its total expenses. Total expenses are defined as the sum of advertising expenses, interest expense, R&D expense, and selling, general, and administrative expense.	Datastream
Diversification	The number of business segments for a firm-year.	Datastream
CF volatility	Three-year standard deviation of operation cash flow	Datastream
Degree of foreign operations		
FNTN (%)	The ratio of the number of foreign subsidiaries to the total number of subsidiaries of a firm.	Orbis
FATA (%)	The ratio of foreign assets to the total assets of a firm.	Datastream
FSTS (%)	The ratio of foreign sales to the total sales of a firm.	Datastream
FITI (%)	The ratio of foreign operating income to the total operating income of a firm.	Datastream
Alternative corruption measures		
Excess regulation	The index of the overabundance of regulation or unnecessary restriction of business activity, which covers 85 major countries in the world and score range from one to five.	Wilhelm (2002)
Black market	The index of black market activity, which covers 85 major countries in the world and score range from one to five.	Wilhelm (2002)
State-level corruption		
Conviction Butler(2009)	the average number of corruption-related convictions per million population according to the number reported in Butler et al. (2009).	Butler et al. (2009)
Conviction Dass(2016)	the average number of corruption-related convictions per million population according to the number reported in Dass et al. (2016).	Dass et al. (2016)
Conviction Smith(2016)	The average number of corruption-related convictions per million population according to the number reported in Smith (2016), who originally uses district-level data. We aggregate this data into the state-level according to the firm-year observations offered by the author.	Smith (2016)

Appendix III: Chinese firms with African operations for matching

Company Name	Business Description
Guizhou Changzhen Tianzheng Holding	Guizhou changzheng tiancheng holding co.,ltd. Mainly operates through three segments, including electrical equipment manufacture segment, financial segment and mineral resource development segment. Electrical equipment manufacture segment is involved in the manufacture and distribution of high-voltage and medium-voltage electrical equipment. Financial segment provides financial services for small and micro merchants. Mineral resource development segment is engaged in the exploration, processing, refining and import and export trading of mineral resources, with zirconium and ilmenite concentrate as its main products. The company distributes its products in domestic market and to overseas markets.
Kunming Pharmaceutical Corporation	Kpc pharmaceuticals, inc. Is a china-based company principally engaged in the research and development, manufacturing and sales of pharmaceuticals. The company's main products include natural herbal medicines, traditional chinese medicines and chemical synthetic drugs, such as central nervous system drugs, cardiovascular and cerebrovascular drugs, anti-parasitic drugs, anti-microbial drugs, anti-gout drugs, hormones and endocrine drugs, hemorrhoids medicines and anti-inflammatory drugs, among others. The company is also engaged in the production of health products and provision of medical services.
Henan Rebecca Hair Products	Henan rebecca hair products co., ltd is a china-based company principally engaged in the manufacture and distribution of hair products. The company's principal products are synthetic hairpieces, chemical fiber hairpieces, chemical fiber wigs, human hair, wigs, artificial heads, and fire-retardant fibers. The company offers its products under the brand named rebecca. It distributes its products in domestic and overseas markets, including the americas, europe, africa, asia and oceania.
Keda Clean Energy Company Limited	Keda clean energy co ltd. Keda clean energy co., ltd. Is a china-based company principally engaged in the manufacture and sales of building material machineries. The company's main products include building ceramic machineries, wall material machineries, stone machineries, hydraulic pumps, compressors, blowers and materials of lithium batteries. The company is also involved in the provision of rental services and the clean energy environmental protection businesses, including the clean coal gasification and flue gas treatment businesses.
Sumec Corporation Limited	Sumec corporation limited is a china-based company mainly engaged in trading business. The company's main business is the import of bulk commodities and mechanical and electrical equipment, garden machinery, gasoline and diesel power generation equipment, auto parts, high-speed rail parts and other mechanical and electrical products and textile and apparel products research and development, production, trade, and new energy projects, ships engineering, environmental engineering, etc. The company distributes its products in domestic market and to overseas markets.
Chengtun Mining Group Company Limited	Chengtun mining group co.ltd is a china-based company, principally engaged in the trading and mining-dressing of nonferrous metal products. The company is engaged in the mining of copper, tin, tungsten, zinc, lead, gold and silver, as well as minerals and non-ferrous metals integrated trading business. The company is also engaged in industrial value-added services business, including geological exploration, evaluation of resources value and inventory management, among others.
Sinoma International Engineering Company Limited	Sinoma international engineering co., ltd is a china-based company, principally engaged in engineering construction business, including cement engineering contracting business and diversified engineering business. The company is also engaged in equipment manufacturing business, environmental protection business and production operation management business. The company's equipment includes crushing equipment, grinding equipment, firing equipment, energy-saving equipment, among others. The company distributes its products within domestic markets and to overseas markets.
China National Chemical Engineering Company Limited	China national chemical engineering co., ltd is principally engaged in engineering construction and design. The company operates its businesses through engineering construction, as well as survey, design and related services. The company's businesses are primarily applied in industries such as chemical, petrochemical, coal chemical, electrical power, construction and environmental protection, among others. It operates its businesses in domestic and overseas markets.
China Railway Construction Corporation Limited	China railway construction corporation limited is a china-based company principally engaged in the engineering contracting businesses. The company's engineering contracting businesses mainly include the construction of railways, highways, urban tracks, water conservancy and hydropower projects, buildings, municipal projects, bridges, tunnels, airports and marine ports, among others. The company is also engaged in the industrial manufacturing, the real estate development, the logistics and materials trading businesses, as well as the provision of survey, design and consulting services. The company mainly operates its businesses in domestic and overseas markets.
Baiyin Non-Ferrous Group	Baiyin nonferrous group co., ltd. Is a china-based company principally engaged in the mining, smelting, processing and trading of nonferrous metals, including copper, lead, zinc, gold and silver. The company's main products include cathode copper, zinc ingots, electric silver, electric lead, gold and others. The company is also engaged in the provision of nonferrous metal processing services. The company distributes its products within domestic market and to overseas markets.
China Railway Group Limited	China railway group limited is a china-based company principally engaged in infrastructure construction business. The company is also engaged in survey, design and consulting services business, the manufacture of engineering equipment and components, and real estate development. Its infrastructure construction business includes railway construction, highway construction and urban engineering construction. Its survey, design and consulting services business includes the provision of survey, design and consulting services for railway, bridge, road and tunnel projects. The company conducts its businesses within in china and overseas markets.
China State Construction Engineering Corporation Limited	China state construction engineering corporation. China state construction engineering corporation limited is a china-based company principally engaged in the engineering contracting businesses. The company's main businesses include the construction contracting of buildings, infrastructure construction and investment, real estate development and investment, electricity, energy exploitation, survey and design, as well the municipal utility design services. The company mainly operates its businesses in domestic and overseas markets.
Power Construction Corporation Of China	Power construction corporation of china ltd. Power construction corporation of china, ltd. Is a china-based company principally engaged in the contracting, survey and design of water conservancy and hydropower projects. The company's principal businesses include the construction contracting, electric investment and operations, real estate development, equipment manufacturing and leasing, among others. The company is also engaged in the commodity trading, the sales of materials and the construction of highways and bridges through its subsidiaries. The company mainly operates its businesses in the domestic and overseas markets.
Petrochina Company Limited	Petrochina company limited is a china-based company principally engaged in the production and distribution of oil and gas. The company mainly operates through four business segments. The exploration and production segment is principally engaged in the exploration, development, production and sales of crude oil and natural gas. The refining and chemical products segment is principally engaged in the refining of crude oil and petroleum products, as well as the production and sales of basic petrochemical products, derivative petrochemical products and other chemical products. The sales segment is principally engaged in the sales of refined petroleum products. The natural gas and pipeline segment is engaged in the transportation and sales of natural gas, crude oil and refined petroleum products.

Zijin Mining Group Company Limited	Zijin mining group company limited is a china-based company principally engaged in the exploration, mining, smelting and sales of gold and other metal minerals. The company's main products include gold products, copper products, lead and zinc products, iron products, silver products, among others. Its gold products mainly include the mineral gold, standard gold bullions, gold ingots and gold concentrates, among others. The company is also involved in the geographic survey businesses through its subsidiaries.
Loncin Motor Company Limited	Loncin motor co ltd. Loncin motor co., ltd. Is a china-based company principally engaged in the manufacturing of various kinds of machineries. The company's primary businesses include the development, production and sales of engines (including road engines and off-road engines), motorcycles, generators (such as small household generators and large commercial generators), micro-electric vehicles, unmanned aerial vehicles and automotive parts, among others. The company mainly operates its businesses in domestic and overseas markets.
Jiayou International Logistics Company Limited	Jiayou international logistics co ltd. Jiayou international logistics co., ltd. Is a china-based company principally engaged in the formulation of cross-border integrated logistics solutions based on customer personalized needs. The company mainly has four business segments. The cross-border multimodal transport segment mainly includes international project logistics and international multimodal transport business. The bulk mineral logistics segment mainly provides full standardization of logistics services. The intelligent storage segment mainly provides intelligent warehousing business services through customs bonded warehouses and customs supervision places at gan qimao port. The supply chain services segment mainly provides the overall market transaction information, customer price feedbacks, logistics solutions, warehousing, settlement and other services.
Jchx Mining Management Company Limited	Jchx mining management co., ltd. Is principally engaged in mine construction engineering and mining operations management businesses. Mine construction engineering business involves in mine capital phase construction engineering, mine production phase reconstruction construction engineering and other single technical improvement measures engineering, among others. The company also involves in design and consulting of mine engineering.
China International Marine Containers (Group)	China international marine containers (group) co., ltd. Is a china-based company principally engaged in the manufacture and distribution of road transport vehicles and containers. The company's main businesses include container manufacturing business; road transport vehicle business; energy, chemical and liquid food equipment business; ocean engineering business; logistics service business; heavy duty truck business; airport equipment business; real estate development business; financial business and others. The company's container products include dry cargo containers, standard reefer containers and specific cargo containers, among others. The company distributes its products in domestic market and to overseas markets.
Norinco International Cooperation Limited	Norinco international cooperation ltd. Is a china-based principally engaged in international engineering contracting business. The company is also involved in various other businesses, including domestic construction engineering, real estates, heavy equipment export trading, logistics services, logistics automation system integration services, solar energy product trading and new energy project development, as well as the manufacture and distribution of metal packaging containers. The company operates its businesses in domestic market and overseas markets.
China National Complete Plant	China national complete plant import & export corporation limited is a china-based company principally engaged in set equipment exporting and project contracting businesses. The company operates through set equipment exporting and project contracting segment, general merchandise trading segment and industrial investment and operation segment. Set equipment exporting and project contracting segment operates its business in various industries, including industrial, electric power, transportation, infrastructure and other industries throughout asia, africa and latin america. General merchandise trading segment mainly exports textiles, electric machinery, solar power systems and agricultural accessories, among others. Industrial investment and operation segment is mainly involved in the production and sale of sucrose and alcohol in overseas market in the form of leasing.
China Nonferrous Metal	China nonferrous metal industry's foreign engineering and construction co.,ltd. Is a china-based company principally engaged in the mining, selecting and smelting of non-ferrous metals. The company's main products include lead concentrates, zinc concentrates, lead ingots and rare earth oxides, among others. The company is also involved in the international project contracting, the manufacture and trading of equipment and other businesses. The company operates its business in domestic and overseas markets.
China Wu Yi	Chinawuyico.,ltd. Is a china-based company principally engaged in the undertaking of construction engineering projects and the development of real estates. The company's construction contracting projects include building construction engineering projects, such as residential buildings, hospitals and convention centers, as well as public infrastructure projects, such as highways, bridges and airports. The company's real estate products include apartments, villas, shopping malls, office building and hotels, among others. The company is also involved in international trading, property management, building materials production and engineering supervision businesses. The company operates its business in domestic and overseas markets.
Xuanhua Construction Machinery	Xuanhua construction machinery co., ltd. Is a china-based company principally engaged in the mining development business, as well as the processing and sales of mineral products. The company's main products include copper, magnetite and vermiculite. The company is also involved in the manufacture and sales of engineering machinery and accessories. The company distributes its products in domestic market and to overseas markets.
China Camc Engineering Company Limited	China camc engineering co., ltd. Is an engineering construction contractor. The company undertakes various complete projects with the content of exporting complete sets of equipment and technology. The company's contract projects consist of industry projects, agriculture projects, water projects, electric power projects, transportation projects and others. In addition, the company involves in the domestic and overseas trading business, as well as asset management business. The company operates its business mainly in asia, south america, africa, europe and china.
Hengbao Company Limited	Hengbao co., ltd. Is a china-based company, principally engaged in the manufacture and distribution of various card products. The company's businesses and products include financial integrated circuit (ic) cards, citizen cards, social security cards, health cards, communication and internet of things connection products, intelligent terminal products and security products, among others. The company is also engaged in the provision of notes, modules and software products. The company conducts its businesses within domestic market and to overseas markets.
Canny Elevator Company Limited	Canny elevator co.,ltd is a china-based company principally engaged in the research, development, production and distribution of elevators. The company's main products include elevators, escalators, moving pavements and related parts and components. The company's products are mainly used in residential buildings, shopping malls, hotels and public facilities, among others. The company also provides installation and maintenance services. The company distributes its products in domestic market and to overseas markets.
Chongqing Sansheng Industrial Company Limited	Chongqing sansheng industrial co.,ltd., formerly chongqing sansheng industrial co., ltd.,is a china-based company principally engaged in the comprehensive utilization and product development of plaster resources and pharmaceutical business. The company operates through two main segments. The construction materials and chemical industry segment is mainly involved in the manufacturing and sales of commercial concretes, water reducing agents, expanding agents and

	sulfuric acids. The company's products are mainly applied in the construction, tunnels, bridges, seaports, roads, chemical fertilizers, chemical engineering, light manufacturing, textiles, steels and others. The pharmaceutical business segment is primarily involved in the production, research and development of raw material medicines, preparation products and intermediate products.
Crrc Corporation Limited	Crrc corporation limited, formerly csr corporation limited, is a china-based company, principally engaged in the manufacture and distribution of railway transportation equipment. The company's main businesses consist of the research and development, design, manufacture, repair, distribution and leasing of railway locomotives, electric multiple units (emus), rapid transit vehicles, engineering machinery, mechanical and electrical equipment, electronic devices and components, electronics and environmental protection equipment. The company is also engaged in the provision of related technical services. The company distributes its products within domestic market and to overseas markets.
Ningxia Orient Tantalum	Ningxia orient tantalum industry co., ltd. Is a china-based company principally engaged in the smelting and processing of precious metals. The company's products portfolio consists of tantalum products, niobium products, tantalum alloys, niobium alloys, energy materials and silicon carbide, among others. The company's products are mainly applied in electronics, metallurgy, chemical engineering, aviation, atomic energy and other fields. The company is also engaged in the import and export trading of nonferrous metals and equipment. The company distributes its products within domestic market and to overseas markets.